

Phase Coexistence in an Eight-Component Q3-Locked Quantum Anharmonic Crystal

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Abstract

We formulate a fixed-spacing three-dimensional quantum anharmonic crystal with eight real components per lattice site, a ferromagnetic spatial coupling, and a non-radial quartic locking interaction on the internal cube graph Q_3 . The source is coupled to the normalized collective direction $u = 8^{-1/2}(1, \dots, 1)$. This manuscript assembles a conditional theorem package, with every imported hypothesis displayed, for the following chain: finite-volume heat traces and Euclidean loop measures; open-box and periodic thermodynamic pressure; compact tempered DLR states; continuous-loop FKG association; spatial reflection positivity and a three-dimensional infrared bound; a volume-uniform collective Duhamel lower bound; and, in an explicit low-temperature sufficient regime, a strict source cusp and two distinct parity-related zero-source DLR states. The proof uses the general-vector existence results of Kozitsky–Pasurek, the finite-dimensional infrared inequality of Froehlich–Simon–Spencer through an explicit hypothesis crosswalk. The needed Griffiths endpoint specialization and finite spectral Falk–Bruch inequality are proved directly. No scalar or radial phase theorem is imported. This is an external-review edition of a conditional manuscript at research tier T0 (R-497). The spatial DLR, infrared, operator and final composition arguments have been expanded, but signed mathematical review, independent literature review, a clean-snapshot replay, and final content freeze remain open. In particular, this paper makes no claim about a ground state gap, real-time KMS dynamics, a continuum limit, a physical vacuum, cosmology, or closure of any TECT sector. This PDF is supplied for critical external review; its production does not discharge those gates or authorize submission.

1 Scope, status, and contributions

This paper concerns one explicitly defined mathematical model. It is not a claim that the model is the microscopic law of nature. The research record for the assembled content is EXP-000780–EXP-000782 and R-497. R-497 is registered as T0, `claim_bearing=false`, and `INTERNAL_REVIEW_ONLY`. The statements below are therefore written as a conditional composition while the model-specific arguments and their source applicability are subjected to line-by-line independent review.

The paper has seven proof blocks.

- (B1) finite-volume Hamiltonians, heat-trace finiteness, and the exact harmonic-reference/path-integral normalization;
- (B2) open-box pressure, periodic pressure, and locally uniform source convergence;
- (B3) tempered Euclidean DLR existence, compactness, and source-tangent limits;
- (B4) continuous-loop FKG association and the collective covariance inequalities;
- (B5) spatial reflection positivity, finite-mesh FSS Gaussian domination, and the nonzero-mode infrared sum in dimension three;
- (B6) a collective translation, form-domain, and Falk–Bruch lower bound;
- (B7) strict cusp and construction of a parity-related DLR pair under an explicit sufficient condition.

The novelty boundary is intentionally narrow. The standard ingredients are the cited Gibbs/DLR, FKG, infrared, and Griffiths tools. The model-specific content is their typed composition for a non-radial eight-component Q_3 interaction, including the exact source normalization and the collective lower bound. No priority or world-first assertion is made.

The following conclusions are explicitly outside the paper:

- no all-parameter phase theorem, and no phase-absence theorem outside the sufficient regime below;
- no assertion that the two states are extremal, pure, clustering, unique, or the complete DLR simplex;
- no common infinite-volume real-time dynamics, algebraic KMS statement, ground-state phase, or spectral gap;
- no spatial lattice-spacing removal, continuum renormalization, physical-vacuum ordering, cosmological interpretation, or Yang–Mills conclusion;
- no closure of TECT sectors A, C, CP1, Pre-A, or the master theorem;
- no submission, upload, or release tag is authorized by this review PDF; see Section 12.

2 The exact Q3LOCK model and conventions

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $V = L^3$, and take even $L \geq 4$ in the thermodynamic sequence. Positive-direction nearest-neighbour bonds are counted once for $L \geq 4$; the finite-volume source record also specifies the parallel-bond multiplicity convention at $L = 2$. A site variable is $q_y = (q_{y,e})_{e \in \{0,1\}^3} \in \mathbb{R}^8$. The internal graph Q_3 has an edge between two binary triples that differ in one coordinate. Write

$$u = 8^{-1/2}(1, \dots, 1), \quad Q_y = (u, q_y), \quad S_y = |q_y|^2,$$

and

$$D_y^{\text{int}} = \sum_{\{e,f\} \in E(Q_3)} (q_{y,e} - q_{y,f})^2.$$

The superscript distinguishes this internal graph quantity from the Duhamel matrix introduced later.

For $m = \chi/\hbar^2 > 0$, $c, g, \lambda > 0$, $r \in \mathbb{R}$, and real collective source h , define

$$\begin{aligned} U_{L,h}(q) = & \sum_{y \in \Lambda_L} \left[\frac{r}{2} S_y + \frac{g}{4} \sum_e q_{y,e}^4 + \frac{\lambda}{4} \sum_{\{e,f\} \in E(Q_3)} (q_{y,e} - q_{y,f})^2 (q_{y,e}^2 + q_{y,f}^2) - h Q_y \right] \\ & + \frac{c}{2} \sum_{\langle y,z \rangle} |q_y - q_z|^2. \end{aligned} \quad (2.1)$$

No radial or $O(8)$ symmetry is assumed. The Hamiltonian on $L^2(\mathbb{R}^{8V}, dq)$ is

$$H_L(h) = -\frac{1}{2m} \sum_{y,e} \partial_{q_{y,e}}^2 + U_{L,h}(q). \quad (2.2)$$

The phase application uses $r < 0$, but the finite-pressure statements allow arbitrary real r .

Put

$$w(q) = \sum_{y \in \Lambda_L} |q_y|^4, \quad Q_L = H^1(\mathbb{R}^{8V}) \cap L^2(\mathbb{R}^{8V}, w(q) dq). \quad (2.3)$$

The following finite-dimensional lemma records the form and truncation facts used below. It is included to distinguish the closed-form argument from the later spatial and thermodynamic limits.

Lemma 2.1 (Finite-volume form and residual truncation). *For $|h| \leq h_0$ the potential in (2.1) obeys*

$$U_{L,h}(q) \geq \frac{g}{128}w(q) - VC_{\text{phys}}, \quad (2.4)$$

$$|U_{L,h}(q)| \leq C_{\text{abs}} + K_{\text{abs}}w(q), \quad (2.5)$$

where one may take

$$\begin{aligned} C_{\text{phys}} &= \frac{4r^2}{g} + \frac{3}{4}h_0^{4/3}(32/g)^{1/3}, \\ C_{\text{abs}} &= V(|r|/2 + h_0 + 6c), \quad K_{\text{abs}} = g/4 + 3\lambda + |r|/8 + h_0/4 + 3c/2, \\ C_{\text{res}}(a) &= \frac{4(r-a)^2}{g} + \frac{3}{4}h_0^{4/3}(32/g)^{1/3}. \end{aligned} \quad (2.6)$$

Consequently the form of $H_L(h)$ with domain $\mathbf{Q}_L$ is closed, semibounded, and has compact embedding into $L^2(\mathbb{R}^{8V})$. If $a > 0$, write $\mathbf{Q}_a = H^1(\mathbb{R}^{8V}) \cap L^2(\mathbb{R}^{8V}, |q|^2 dq)$ for the harmonic form domain. For $R = R_{a,h}$ and $R_M = \min(R, M)$ with $M \geq -VC_{\text{res}}(a)$, the forms

$$\mathbf{q}_M[\psi] = \mathbf{q}_a[\psi] + \int R_M |\psi|^2 dq$$

are closed on the harmonic form domain, increase to the physical form on $\mathbf{Q}_L$, and their resolvents converge strongly to the operator associated with $H_L(h)$. The last statement is a finite-volume variational limit; it does not assert a spatial or continuum form limit.

Proof. For a single site, Cauchy–Schwarz gives $\sum_e q_{y,e}^4 \geq |q_y|^4/8$. The locking and spatial difference terms are nonnegative. With $t = |q_y|$, the two scalar maximisations

$$bt^2 - \frac{g}{64}t^4 \leq \frac{16b^2}{g}, \quad h_0t - \frac{g}{128}t^4 \leq \frac{3}{4}h_0^{4/3}(32/g)^{1/3}$$

hold for every $b, t \geq 0$. Taking $b = |r|/2$ gives (2.4); taking $b = |r-a|/2$ gives the residual bound used in (3.2).

For the upper envelope use $(x-y)^2(x^2+y^2) \leq 4(x^4+y^4)$, $t^2 \leq 1+t^4/4$, and $t \leq 1+t^4/4$. Every Q3 vertex has degree three, and every periodic spatial site has six endpoint incidences. Hence the locking term contributes at most $3\lambda w$, the spatial term contributes at most $6cV + (3c/2)w$, and the quadratic and source terms give the remaining parts of (2.6). This proves (2.5).

Choose C_0 so that $U_{L,h} + C_0 \geq 1 + (g/256)w$. The form norm of $\mathbf{q}_h + C_0 \|\cdot\|_2^2$ is then equivalent, by the two displayed envelopes, to

$$\|\nabla\psi\|_2^2 + \|\psi\|_2^2 + \|w^{1/2}\psi\|_2^2.$$

The intersection defining $\mathbf{Q}_L$ is therefore complete. For a form-bounded sequence, Rellich compactness on a ball gives an L^2 -convergent subsequence there, while

$$\int_{|q|>R} |\psi(q)|^2 dq \leq VR^{-4} \int w(q)|\psi(q)|^2 dq$$

controls the tail uniformly. Thus the embedding $\mathbf{Q}_L \hookrightarrow L^2$ is compact.

The residual lower bound gives $H_L(h) \geq H_a - VC_{\text{res}}(a)$ as a form inequality, with $C_{\text{res}}(a)$ from (3.2). For $M \geq -VC_{\text{res}}(a)$, all forms have the harmonic form domain, R_M is continuous and bounded, and $R_M \uparrow R$. Monotone convergence of the positive part of R_M and the common lower bound show $\mathbf{q}_M[\psi] \uparrow \mathbf{q}_h[\psi]$ for $\psi \in \mathbf{Q}_L$. To see the associated resolvent limit directly, fix $\alpha > VC_{\text{res}}(a)$ and let u_M minimize

$$\mathbf{q}_M[u] + \alpha \|u\|_2^2 - 2\Re \langle f, u \rangle.$$

The inequality $\mathbf{q}_M \geq \mathbf{q}_a - VC_{\text{res}}(a) \|\cdot\|_2^2$ and the minimizing inequality give a uniform bound in the harmonic form norm

$$\mathbf{q}_a[u_M] + (\alpha - VC_{\text{res}}(a)) \|u_M\|_2^2.$$

This is deliberately only a harmonic-form bound: the upper truncation does not provide a uniform quartic bound. The harmonic form embedding is compact in finite dimension, so every subsequence has a further subsequence converging weakly in $\mathbf{Q}_a$ and strongly in L^2 to some u . For each fixed $k \geq -VC_{\text{res}}(a)$ and all $M \geq k$, monotonicity gives $\mathbf{q}_k[u_M] \leq \mathbf{q}_M[u_M]$. Since $\mathbf{q}_k$ is closed and boundedly equivalent to the harmonic form norm, weak lower semicontinuity gives

$$\mathbf{q}_k[u] + \alpha \|u\|_2^2 \leq \liminf_{M \rightarrow \infty} (\mathbf{q}_k[u_M] + \alpha \|u_M\|_2^2).$$

The uniform bound and monotone convergence in k imply $\mathbf{q}_h[u] < \infty$, so $u \in \mathbf{Q}_L$. For $F_M(v) = \mathbf{q}_M[v] + \alpha \|v\|_2^2 - 2\Re \langle f, v \rangle$ and any $v \in \mathbf{Q}_L$, minimality gives $F_M(u_M) \leq F_M(v)$. Taking the limsup on the right gives $F_h(v)$. On the other hand, for every fixed k the lower bound $F_M(u_M) \geq \mathbf{q}_k[u_M] + \alpha \|u_M\|_2^2 - 2\Re \langle f, u_M \rangle$ and the preceding lower-semicontinuity estimate give, after $k \rightarrow \infty$, $F_h(u) \leq \liminf_M F_M(u_M)$. Hence u minimizes F_h on $\mathbf{Q}_L$. The minimizer is unique, and compactness of the harmonic embedding then upgrades subsequential convergence to strong L^2 convergence of the full resolvent sequence. This proves the stated resolvent convergence without assuming an operator-monotonicity property of the exponential or a false uniform quartic bound for the upper-truncated forms. $\square$

The lemma supplies only finite-volume operator control. In particular, its compact embedding is not the tempered spatial compactness used in the DLR construction, and the resolvent argument does not by itself identify the unbounded Feynman–Kac kernel; that identification is carried out next by the bounded truncations and Simon’s cited bridge theorem.

Remark 2.2 (Source and units). The parameter h is an energy coefficient in $H_L(h) = H_L(0) - h \sum_y Q_y$. It is not βh . This convention is responsible for every factor of β in the pressure and Duhamel identities below.

3 Finite volume: trace, loops, and harmonic split

Choose an auxiliary $a > 0$, set $\omega_a = \sqrt{a/m}$, and write

$$H_a = -\frac{1}{2m}\Delta + \frac{a}{2} \sum_y |q_y|^2, \quad R_{a,h} = U_{L,h} - \frac{a}{2} \sum_y |q_y|^2. \quad (3.1)$$

For $|h| \leq h_0$, put $b_a = |r - a|/2$ and

$$A = \frac{g}{128}, \quad C = \frac{16b_a^2}{g} + \frac{3}{4}h_0^{4/3}(32/g)^{1/3}, \quad R_{a,h}(q) \geq A \sum_y |q_y|^4 - VC. \quad (3.2)$$

Indeed $\sum_e q_e^4 \geq |q|^4/8$, the locking and spatial differences are nonnegative, and $b_a t^2 \leq gt^4/64 + 16b_a^2/g$ and $h_0 t \leq gt^4/128 + (3/4)h_0^{4/3}(32/g)^{1/3}$ for $t \geq 0$. Min–max comparison with the harmonic oscillator therefore yields

$$0 < Z_L(h) := \text{Tr } e^{-\beta H_L(h)} \leq e^{\beta VC} [2 \sinh(\beta \omega_a/2)]^{-8V} < \infty. \quad (3.3)$$

This is an eigenvalue comparison, not an appeal to operator monotonicity of the exponential.

For an even time mesh $N \geq 4$, $\varepsilon = \beta/N$, and cyclic variables $x_{y,k}$, the Gaussian reference action is

$$S_{a,N}(x) = \frac{1}{2} \sum_{y,k} \left[\frac{m}{\varepsilon} |x_{y,k+1} - x_{y,k}|^2 + a\varepsilon |x_{y,k}|^2 \right]. \quad (3.4)$$

Let $n_L = 8V$, and define the absolute Gaussian prefactor and probability law by

$$c_N = (m/(2\pi\varepsilon))^{n_L N/2}, \quad Z_{a,N} = c_N \int e^{-S_{a,N}} dx, \quad \gamma_{a,N,L}(dx) = \frac{c_N}{Z_{a,N}} e^{-S_{a,N}(x)} dx.$$

The factor c_N is the product of free heat-kernel normalizations. For one coordinate, Gaussian integration cancels it against the precision factor $(m/\varepsilon)^{N/2}$ and leaves the inverse square root of the cyclic determinant with diagonal $2 + t$, off-diagonal -1 , and $t = \varepsilon^2 a/m$. The exact cyclic determinant is

$$Z_{a,N} = \left[2 \sinh \left(N \operatorname{arsinh} \frac{\beta \omega_a}{2N} \right) \right]^{-8V}, \quad Z_{a,N} \longrightarrow [2 \sinh(\beta \omega_a/2)]^{-8V}. \quad (3.5)$$

To see this, put $z + z^{-1} = 2 + t$, $z > 1$. Each Fourier eigenvalue is $(z - \zeta)(z - \zeta^{-1})/z$, with $\zeta^N = 1$. Their product is $z^N + z^{-N} - 2 = 4 \sinh^2(N \log z/2)$ and $\log z = 2 \operatorname{arsinh}(\sqrt{t}/2)$, proving (3.5). The independent recurrence is $T_0 = 2$, $T_1 = 2 + t$, $T_j = (2 + t)T_{j-1} - T_{j-2}$, with determinant $T_N - 2$.

3.1 Gaussian convergence in the finite-volume loop space

All limits in this subsection keep L, β, m, a fixed. Write $\mathbf{E}_L = C_{\text{per}}([0, \beta]; \mathbb{R}^{n_L})$, with its sup norm, and let $\mathcal{I}_N$ be periodic linear interpolation. For one scalar coordinate, unitary discrete Fourier inversion gives the grid covariance

$$\begin{aligned} G_N(j) &= \frac{1}{\beta} \sum_{n \in \mathcal{J}_N} \frac{e^{2\pi i n j / N}}{D_N(n)}, \quad \mathcal{J}_N = \{-N/2 + 1, \dots, N/2\}, \\ D_N(n) &= a + \frac{4m}{\varepsilon^2} \sin^2(\pi n / N), \quad D(n) = a + \frac{4\pi^2 m n^2}{\beta^2}, \\ G(t) &= \frac{1}{\beta} \sum_{n \in \mathbb{Z}} \frac{e^{2\pi i n t / \beta}}{D(n)}. \end{aligned} \quad (3.6)$$

The continuum series is absolutely summable. It is positive definite and defines the finite-dimensional distributions of a centered periodic Gaussian. The increment estimate below provides a continuous version. Both diagonal covariances are bounded by

$$G_N(0), G(0) \leq K_a := \frac{1}{\beta a} + \frac{\beta}{12m}. \quad (3.7)$$

For the discrete bound, keep the zero Fourier mode and remove a only from the remaining denominators. The cycle Laplacian pseudoinverse has diagonal $(N^2 - 1)/(12N)$: its effective resistance at separation j is $j(N - j)/N$, whose sum is twice N times that diagonal. Multiplication by ε/m gives the bound. In the continuum, sum the nonzero comparison coefficients $\beta/(4\pi^2 m n^2)$ to obtain the same upper bound.

For $0 < |n| \leq N/2$, the inequalities $\sin x \geq 2x/\pi$ and $0 \leq x^2 - \sin^2 x \leq x^4/3$ on $[0, \pi/2]$ imply

$$0 \leq D_N(n)^{-1} - D(n)^{-1} \leq \frac{\pi^2 \beta^2}{48mN^2}.$$

The omitted modes obey $\sum_{n \notin \mathcal{J}_N} n^{-2} \leq 6/N$; one may sum the two decreasing tails by integrals and retain the missing negative Nyquist mode separately. Consequently

$$\sup_j |G_N(j) - G(\varepsilon j)| \leq B_N := \frac{\beta}{mN} \left(\frac{\pi^2}{48} + \frac{3}{2\pi^2} \right). \quad (3.8)$$

Let d denote circular distance between two times. The continuum massless comparison, summing the Fourier series for the periodic bridge increment, gives $\mathbb{E}|X(t) - X(s)|^2 \leq d(\beta - d)/(m\beta) \leq d/m$. Cauchy-Schwarz then gives $|G(t) - G(s)| \leq \sqrt{K_a d/m}$. The interpolated covariance is a convex combination of four grid covariances, and the corresponding time differences are within 2ε of $t - s$. Therefore

$$\sup_{s,t} |\operatorname{Cov}(\mathcal{I}_N X(t), \mathcal{I}_N X(s)) - G(t - s)| \leq B_N + \sqrt{\frac{2K_a \beta}{mN}} \longrightarrow 0. \quad (3.9)$$

This proves convergence of every finite Gaussian evaluation vector.

Tightness requires an estimate at arbitrary times, not just grid vertices. For a grid increment at shortest separation j , comparison on the zero-sum subspace with the massless cycle gives $\mathbb{E}|x_k - x_l|^2 \leq \varepsilon j/m$. If $d \leq \varepsilon$, the interpolated increment crosses at most two partial edges; the L^2 triangle inequality bounds its variance by $d^2/(m\varepsilon) \leq d/m$. If $d > \varepsilon$, insert neighboring vertices at both ends. The two end increments have L^2 norms at most $\sqrt{\varepsilon/m}$, and the intervening grid increment has norm at most $\sqrt{(d+2\varepsilon)/m}$. It follows that

$$\begin{aligned} \mathbb{E}|\mathcal{I}_N X(t) - \mathcal{I}_N X(s)|^2 &\leq 14d/m, \\ \mathbb{E}\|\mathcal{I}_N X(t) - \mathcal{I}_N X(s)\|^4 &\leq n_L(n_L + 2)14^2 d^2/m^2. \end{aligned} \quad (3.10)$$

Here $(2+\sqrt{3})^2 < 14$, and the second line uses independence of the n_L coordinates and the exact fourth moment of a centered Gaussian. Kolmogorov's tightness criterion, with fourth-moment exponent $2 > 1$ and the tight one-time marginals from (3.7), applies on the time interval. Periodicity is a closed condition. Tightness and (3.9) thus prove

$$\nu_N := (\mathcal{I}_N)_* \gamma_{a,N,L} \implies \gamma_{a,L} \quad \text{on } \mathbf{E}_L. \quad (3.11)$$

Every subsequential limit has the same Gaussian finite-dimensional laws, which determine a measure on this continuous-path space. This proves convergence of the whole sequence. No spatial weighted topology or uniformity as $L \rightarrow \infty$ has been used.

3.2 Interacting weights, normalizers and source moments

The full mesh action is independent of a after cancellation:

$$S_{a,N}(x) + \varepsilon \sum_k R_{a,h}(x_k) = \frac{m}{2\varepsilon} \sum_{y,k} |x_{y,k+1} - x_{y,k}|^2 + \varepsilon \sum_k U_{L,h}(x_k). \quad (3.12)$$

Define on all of $\mathbf{E}_L$

$$\mathcal{R}_{N,h}(\omega) = \varepsilon \sum_k R_{a,h}(\omega(k\varepsilon)), \quad \mathcal{R}_h(\omega) = \int_0^\beta R_{a,h}(\omega(t)) dt, \quad w_{N,h} = e^{-\mathcal{R}_{N,h}}, \quad w_h = e^{-\mathcal{R}_h}.$$

These expressions agree with the mesh residual on polygonal paths. Equation (3.2) bounds both weights above by $M_L = e^{\beta VC}$. Under either Gaussian reference, Wick averaging gives the uniform upper bound $\mathbb{E}\mathcal{R}_{N,h}, \mathbb{E}\mathcal{R}_h \leq \beta VC_J^{\text{mesh}}$, where

$$C_J^{\text{mesh}} = 4|r-a|K_a + 24cK_a + (6g+24\lambda)K_a^2.$$

Indeed a scalar fourth moment is $3s^2$, an internal edge polynomial has mean $8s^2$, there are twelve internal edges and at most three spatial bonds per site, and $s \leq K_a$. The source has zero Gaussian mean. Jensen's inequality consequently gives

$$\mathfrak{z}_{N,h} := \int w_{N,h} d\nu_N \geq e^{-\beta VC_J^{\text{mesh}}}, \quad \mathfrak{z}_h := \int w_h d\gamma_{a,L} \geq e^{-\beta VC_J^{\text{mesh}}}. \quad (3.13)$$

The constants are uniform in N and $|h| \leq h_0$ at this fixed volume.

Let $\mathcal{K} \subset \mathbf{E}_L$ be compact. Its paths have bounded range and a common modulus of continuity $\eta_{\mathcal{K}}$ tending to zero. The finite-dimensional polynomial $R_{a,h}$ has a common Lipschitz constant $L_{\mathcal{K}}$ on that range for $|h| \leq h_0$. Thus

$$\sup_{\omega \in \mathcal{K}, |h| \leq h_0} |\mathcal{R}_{N,h}(\omega) - \mathcal{R}_h(\omega)| \leq \beta L_{\mathcal{K}} \eta_{\mathcal{K}}(\varepsilon) \longrightarrow 0. \quad (3.14)$$

The kinetic action is already included in ν_N ; it is not evaluated as a Riemann sum of a derivative of a typical Gaussian path.

For any bounded continuous F , choose a common compact set carrying all but probability η of every ν_N , using (3.11). On this set the weights converge uniformly by (3.14); on its complement,

$|F(w_{N,h} - w_h)| \leq 2\|F\|_\infty M_L$. Finally Fw_h is bounded continuous, so ordinary weak convergence applies to it. Taking $N \rightarrow \infty$ and then $\eta \downarrow 0$ proves

$$\int Fw_{N,h} d\nu_N \longrightarrow \int Fw_h d\gamma_{a,L}, \quad \mathfrak{z}_{N,h} \longrightarrow \mathfrak{z}_h > 0, \quad \mu_{N,L,h} := \frac{w_{N,h}\nu_N}{\mathfrak{z}_{N,h}} \Longrightarrow \mu_{L,h} := \frac{w_h\gamma_{a,L}}{\mathfrak{z}_h}. \quad (3.15)$$

This is weak convergence on $\mathbf{E}_L$, not total-variation convergence. The absolute mesh partition function is $Z_{N,L}(h) = Z_{a,N}\mathfrak{z}_{N,h}$, so it converges to $Z_a\mathfrak{z}_h$. The full finite-mesh action (3.12) and its prefactor c_N do not depend on a . Repeating the limit for any other fixed positive split therefore proves independence of both this absolute limit and the normalized loop law from the chosen split.

3.3 Identification with the heat trace

The precise external input is Simon [1, Theorem 1.1, (1.1)–(1.3), page 2]: the Brownian-bridge formula for a continuous potential bounded below on a finite-dimensional Euclidean space. Its kinetic operator is $-\Delta/2$. With $D = 8|\Lambda_L|$ and $z = \sqrt{m}q$, the unitary map $\psi(q) \mapsto m^{-D/4}\psi(z/\sqrt{m})$ sends our operator to $-\Delta_z/2 + U_{L,h}(z/\sqrt{m})$. The polynomial $U_{L,h}$ is continuous and bounded below at every fixed L, h , by the retained quartic bound. In the original coordinates the free bridge has covariance $m^{-1}(\min(s, t) - st/\beta)$, and its kernel prefactor is $(m/(2\pi\beta))^{D/2}$. No mass-one convention is silently imposed. The stronger unbounded-semigroup Theorem 1.2 of that source is not used. Its weaker growth condition (1.4) is therefore not being substituted for the bounded-below hypothesis in this finite-volume application.

The harmonic-reference version follows from this free-bridge input by defining

$$d\mathbb{P}_{q,q'}^a = \frac{\exp[-\int_0^\beta a|\omega(t)|^2/2 dt]}{\mathbb{E}_{q,q'}^0 \exp[-\int_0^\beta a|\omega(t)|^2/2 dt]} d\mathbb{P}_{q,q'}^0. \quad (3.16)$$

The denominator is strictly positive. Applying the free formula to the harmonic potential alone identifies it with K_a/K_0 . Applying the same formula to $U_{L,h} = a|q|^2/2 + R_{a,h}$ gives the following identity. For the positive harmonic operator H_a , a continuous real multiplication potential R bounded below, and $H = H_a + R$ defined by its semibounded form, its kernel is

$$K_H(\beta; q, q') = K_a(\beta; q, q') \mathbb{E}_{q,q'}^a \exp \left[-\int_0^\beta R(\omega(t)) dt \right].$$

Here $\mathbb{E}_{q,q'}^a$ denotes the normalized harmonic bridge. The bounded potential formula extends to R by truncating R above and passing by monotone form convergence; the bridge factors are dominated by $e^{-\beta \inf R}$. The kernels are continuous by local endpoint dominated convergence in the bridge formula. Their semigroup property and symmetry give $\int K_H(\beta; q, q) dq = \|e^{-\beta H/2}\|_{\text{HS}}^2 = \text{Tr } e^{-\beta H}$ by Tonelli; finiteness follows also from the integrable majorant $e^{-\beta \inf R} K_a(\beta; q, q)$. Mixing the diagonal bridges with $K_a(\beta; q, q) dq/Z_a$ gives the centered periodic Gaussian with covariance $(-m\partial_t^2 + a)^{-1}$ in (3.6). Applying this formula with $R = R_{a,h}$ gives

$$Z_L(h) = Z_a \mathbb{E}_{\gamma_{a,L}} \exp \left[-\int_0^\beta R_{a,h}(\omega(\tau)) d\tau \right], \quad (3.17)$$

The parameter and form hypotheses are verified in (2.3) and (3.2). The source-specific normalized Feynman–Kac formulas being compared are KP [2], Section 2.3, equations (2.28)–(2.32). Their Z_Λ in (2.32) is the residual normalizer $\mathfrak{z}_h$; the absolute heat trace here includes Z_a . This identifies the weak limit above with the operator loop law and supplies the absolute normalization without inferring it from a normalized correlation identity. Simon’s free-bridge theorem is the external analytic input; the mass scaling, harmonic reweighting and trace normalization above are its finite-volume application, not additional infinite-volume imports.

3.4 Unbounded sources and pressure derivatives

For a fixed real site field f , define

$$Y_f(\omega) = \int_0^\beta \sum_y f_y(u, \omega_y(t)) dt, \quad Y_{N,f}(\omega) = \varepsilon \sum_{y,k} f_y(u, \omega_y(k\varepsilon)), \quad \mathcal{Q}_N(\omega) = \varepsilon \sum_{y,k} |\omega_y(k\varepsilon)|^4.$$

On polygonal paths $Y_f = Y_{N,f}$ exactly: the integral of each affine segment is the average of its endpoints, and cyclic summation counts each endpoint twice. Holder's inequality gives $|Y_{N,f}| \leq B_f \mathcal{Q}_N^{1/4}$, where $B_f = (\beta \sum_y |f_y|^{4/3})^{3/4}$. For $b, \delta > 0$, maximization over $z \geq 0$ gives

$$bz - \delta z^4 \leq C_Y(b, \delta) := \frac{3b^{4/3}}{4(4\delta)^{1/3}}.$$

Using (3.2) and (3.13), for every $T > 0$, $p > 1$ and $0 < \delta < A$, we obtain

$$\sup_{N, |h| \leq h_0} \mathbb{E}_{\mu_{N,L,h}} e^{pT|Y_{N,f}|} \leq \exp\{\beta V(C + C_J^{\text{mesh}}) + C_Y(pTB_f, \delta)\}. \quad (3.18)$$

The analogous continuous estimate follows by integral Holder and the same Young inequality. A slightly larger T absorbs any fixed power of $|Y_{N,f}|$. Truncation and (3.15) therefore pass the source exponential and its first two derivatives to the limit.

Point evaluations use a different bound. The density relative to the Gaussian reference is at most $e^{\beta V(C + C_J^{\text{mesh}})}$, and a polygonal Gaussian evaluation has variance at most K_a . Hence

$$\sup_{N, |h| \leq h_0, t, y, e} \mathbb{E}_{\mu_{N,L,h}} |\omega_{y,e}(t)|^4 \leq 3K_a^2 e^{\beta V(C + C_J^{\text{mesh}})}. \quad (3.19)$$

The same bound holds for $\mu_{L,h}$. It is not a spatially uniform bound; the later DLR passage needs its own estimate. In particular an integrated quartic norm has not been used to bound a point evaluation deterministically.

Define

$$p_{\beta,L}(h) = \frac{1}{V} \log Z_L(h), \quad P_{\beta,L}(h) = \frac{p_{\beta,L}(h)}{8\beta}, \quad X_L = \int_0^\beta \sum_y Q_y(\tau) d\tau. \quad (3.20)$$

Quartic domination implies $\mathbb{E}_{L,0} e^{T|X_L|} < \infty$ for every finite T . Indeed, if $\mathcal{Q} = \int_0^\beta \sum_y |\omega_y|^4 d\tau$, then $|X_L| \leq (\beta V)^{3/4} \mathcal{Q}^{1/4}$, and the retained positive quartic term absorbs $T|X_L|$ by Young's inequality. Thus $Z_L(z) = Z_L(0) \mathbb{E}_{L,0} e^{zX_L}$ extends to an entire function of the complex source z : on $|z| \leq T$, the integrand and each of its fixed-order derivatives have a common integrable exponential majorant. Since $Z_L(h) > 0$ for every real h , it has a nonvanishing complex neighborhood of each such point, on which a holomorphic logarithm exists. It follows that $p_{\beta,L}$ is real analytic on $\mathbb{R}$. An entire continuation of $\log Z_L$ would require exclusion of complex zeros and is not asserted. For real h ,

$$\begin{aligned} p'_{\beta,L}(h) &= \frac{\mathbb{E}_{L,h} X_L}{V} = \beta \mathbb{E}_{L,h} Q_0(0), & P'_{\beta,L}(h) &= \frac{\mathbb{E}_{L,h} Q_0(0)}{8}, \\ p''_{\beta,L}(h) &= \frac{\text{Var}_{L,h}(X_L)}{V} \geq 0. \end{aligned} \quad (3.21)$$

At $h = 0$, parity $q \mapsto -q$ gives $p'_{\beta,L}(0) = 0$. A cusp can therefore occur only after the spatial limit.

4 Thermodynamic pressure

Let $\Lambda \subset \mathbb{Z}^3$ be an even rectangle. The open Hamiltonian keeps only bonds with both endpoints in Λ ; the periodic Hamiltonian on an even cube keeps the full positive-direction bond multiset. The same onsite polynomial and source are used in both cases.

Here are the uniform estimates needed for the limit; a finite upper bound alone would not exclude a limit of minus infinity. Put $r_- = \max(-r, 0)$. For $|h| \leq H$, scalar Young inequalities give, as quadratic forms,

$$H_\Lambda^{\text{op}}(h) \geq Q_\Lambda - b(H)|\Lambda|, \quad Q_\Lambda = \frac{g}{8} \sum_{y,e} q_{y,e}^4, \quad b(H) = \frac{8r_-^2}{g} + \frac{3}{2}(4/g)^{1/3} H^{4/3}. \quad (4.1)$$

To verify the constants, retain $gx^4/8$ from each scalar quartic and use $gx^4/16 - r_-x^2/2 \geq -r_-^2/g$ and $gx^4/16 - |J_e||x| \geq -(3/4)(4/g)^{1/3}|J_e|^{4/3}$, where $J_e = h/\sqrt{8}$ and $\sum_e |J_e|^{4/3} = 2|h|^{4/3}$.

For $\beta \in [b_0, b_1] \subset (0, \infty)$, the variance of one harmonic reference coordinate is

$$k_a(\beta) = \frac{1}{2m\omega_a} \coth(\beta\omega_a/2) \leq K := \frac{1}{b_0a} + \frac{b_1}{12m}.$$

The inequality follows from $\coth x \leq x^{-1} + x/3$ for $x > 0$; for example the partial-fraction expansion of $\coth$ bounds the remainder by $2x \sum_{n \geq 1} (\pi n)^{-2} = x/3$. The scalar quartic and each internal edge have Gaussian means $3k_a^2$ and $8k_a^2$, respectively. There are eight coordinates and twelve internal edges per site, and at most three spatial bonds per site. The centered source has mean zero. Consequently, for either boundary condition,

$$\begin{aligned} C_J &= 4|r-a|K + 24cK + (6g + 24\lambda)K^2, \\ C &= \frac{4(r-a)^2}{g} + \frac{3}{4}(32/g)^{1/3} H^{4/3}, \\ -8 \log[2 \sinh(\beta\omega_a/2)] - \beta C_J &\leq |\Lambda|^{-1} \log Z_\Lambda(\beta, h) \leq -8 \log[2 \sinh(\beta\omega_a/2)] + \beta C. \end{aligned} \quad (4.2)$$

The lower bound is Jensen's inequality in (3.17); the upper bound uses $R_{a,h} \geq -C|\Lambda|$, obtained by reserving half the quartic term and absorbing the quadratic and source terms in the other half. All constants are independent of Λ . In particular the absolute value of every normalized log trace is bounded by some $M < \infty$ on the displayed compact parameter rectangle.

Proposition 4.1 (Open-box pressure from the finite-volume representation). *For fixed $\beta > 0$ and $h \in \mathbb{R}$, the limit*

$$p_\beta^{\text{op}}(h) = \lim_{\Lambda \nearrow \mathbb{Z}^3} |\Lambda|^{-1} \log Z_\Lambda^{\text{op}}(h) \quad (4.3)$$

exists along arbitrary even rectangular exhaustions and equals the infimum of the normalized open-box values. It is finite, even, and convex in h .

Proof. Cutting a rectangle into two rectangles deletes nonnegative crossing bonds. Min-max comparison and tensor-product trace factorization give $F(\Lambda) = \log Z_\Lambda^{\text{op}} \leq F(\Lambda_1) + F(\Lambda_2)$. Fix an even block B with side lengths b_i and tile a rectangle with even side lengths l_i by copies of B and even remainder rectangles. If v_{rem} is the remainder volume, then

$$\frac{v_{\text{rem}}}{|\Lambda|} = 1 - \prod_{i=1}^3 \left(1 - \frac{l_i \bmod b_i}{l_i}\right) \leq \sum_{i=1}^3 \frac{l_i \bmod b_i}{l_i} \rightarrow 0.$$

Subadditivity and (4.2) bound $F(\Lambda)/|\Lambda|$ above by $(1 - v_{\text{rem}}/|\Lambda|)F(B)/|B| + Mv_{\text{rem}}/|\Lambda|$. Therefore its limsup is at most $F(B)/|B|$ for every B . Every term is at least $\inf_B F(B)/|B|$, which is finite by the lower bound in (4.2). This proves the asserted limit and infimum formula. Convexity and evenness pass from finite volume to this finite limit. $\square$

For a periodic cube, write $B_L^{\text{seam}} = H_L^{\text{per}}(h) - H_L^{\text{op}}(h)$ for the multiplication form given by the added seam bonds. It is distinct from the collective displacement Hessian B_L in Section 8. The seam contains $3L^2$ bonds and $48L^2$ scalar endpoint occurrences. The endpoint Young estimate, with an arbitrary $\eta > 0$, gives the corrected bound

$$0 \leq B_L^{\text{seam}} \leq \eta Q_L + \frac{288c^2L^2}{\eta g}, \quad Q_L = \frac{g}{8} \sum_{y,e} q_{y,e}^4. \quad (4.4)$$

For clarity, there are at most three seam endpoint occurrences at any scalar coordinate, and $cx^2 \leq (\eta g/24)x^4 + 6c^2/(\eta g)$. Thus the coefficient in (4.4) is $48 \times 6 = 288$. Combining this bound with (4.1) on the common form domain gives

$$\begin{aligned} H_L^{\text{op}} &\leq H_L^{\text{per}} \leq (1 + \eta)H_L^{\text{op}} + D_L^{\text{seam}}, \\ D_L^{\text{seam}} &= \eta b(H)L^3 + \frac{288c^2L^2}{\eta g}, \\ e^{-\beta D_L^{\text{seam}}} Z_L^{\text{op}}(\beta(1 + \eta), h) &\leq Z_L^{\text{per}}(\beta, h) \leq Z_L^{\text{op}}(\beta, h). \end{aligned} \quad (4.5)$$

The last line follows from min-max eigenvalue comparison and summation.

Write $f_L(\beta, h) = L^{-3} \log Z_L^{\text{op}}(\beta, h)$. On $[\beta_0/2, 2\beta_1] \times [-H - 1, H + 1]$, choose the uniform absolute bound M supplied by (4.2). For fixed h , f_L is convex in β (its second derivative is the Gibbs energy variance per site). Endpoint secants therefore bound its slopes on $[\beta_0, 3\beta_1/2]$ by $4M/\beta_0$ in absolute value. For $0 < \eta \leq 1/2$ and $\beta \in [\beta_0, \beta_1]$ this gives

$$|f_L(\beta(1 + \eta), h) - f_L(\beta, h)| \leq (4M\beta_1/\beta_0)\eta.$$

Taking $\eta = L^{-1/2}$ in (4.5) yields

$$0 \leq f_L(\beta, h) - p_{\beta, L}^{\text{per}}(h) \leq \left(\frac{4M\beta_1}{\beta_0} + \beta_1 b(H) + \frac{288\beta_1 c^2}{g} \right) L^{-1/2}. \quad (4.6)$$

Endpoint secants in h also give a uniform slope bound on $[-H, H]$. Comparing two parameter points one coordinate at a time proves common local Lipschitz bounds. A finite net upgrades the pointwise open-box limit to local uniform convergence; (4.6) transfers it to periodic cubes. Separate convexity suffices; joint convexity in (β, h) is not needed. Thus

$$p_\beta(h) = \lim_{L \rightarrow \infty, L \text{ even}} p_{\beta, L}^{\text{per}}(h) = p_\beta^{\text{op}}(h). \quad (4.7)$$

The limit is locally uniform in (β, h) , finite, even, and convex in h . At every differentiability point of $P_\beta(h) = p_\beta(h)/(8\beta)$,

$$P'_{\beta, L}(h) \longrightarrow P'_\beta(h). \quad (4.8)$$

Indeed, at fixed β and any sufficiently small $a_1 > 0$, convexity gives

$$\frac{P_{\beta, L}(h) - P_{\beta, L}(h - a_1)}{a_1} \leq P'_{\beta, L}(h) \leq \frac{P_{\beta, L}(h + a_1) - P_{\beta, L}(h)}{a_1}.$$

First take the volume limit in the two secants, then let $a_1 \downarrow 0$. Differentiability of the limiting pressure at this fixed h squeezes both bounds to the same derivative. No differentiability at $h = 0$ is assumed.

5 Tempered DLR states and source tangents

Let $C_\beta = C_{\text{per}}([0, \beta]; \mathbb{R}^8)$ and $\Omega = \prod_{y \in \mathbb{Z}^3} C_\beta$. For $\alpha > 0$ use

$$\|\omega\|_\alpha^2 = \sum_y e^{-\alpha|y|} \|\omega_y\|_{L^2}^2, \quad \Omega_t = \bigcap_{\alpha > 0} \{\|\omega\|_\alpha < \infty\}, \quad (5.1)$$

where $|y|$ is Euclidean distance. Put $\alpha_k = 1/k$ and enumerate the sites as $(y_j)_{j \geq 1}$. The topology used throughout this section is metrized by

$$d_t(\omega, \eta) = \sum_{k \geq 1} 2^{-k} (1 \wedge \|\omega - \eta\|_{\alpha_k}) + \sum_{j \geq 1} 2^{-j} (1 \wedge \|\omega_{y_j} - \eta_{y_j}\|_\infty). \quad (5.2)$$

The sequence $\alpha_k \downarrow 0$ is cofinal for (5.1). This is a Polish space: a Cauchy sequence has compatible limits in every weighted L^2 space and in every local continuous loop space, and these limits define an element of Ω_t . Finite-support loops with rational polygonal values at rational fractions of β form a countable dense set. Weighted norms are measurable in the product sigma-field by their finite partial sums; local sup norms are measurable by countable time evaluations. Thus the Borel sigma-field of this topology is the trace of the product sigma-field. Write W_t for weak convergence of probability measures for this topology. Fix $0 < \sigma < 1/2$ and use the periodic C^σ norm, including the sup norm, for compactness estimates.

5.1 Specification and the general-vector input

Choose $a > 0$ and the normalized oscillator Gaussian χ_a . The allocated one-site potential and pair interaction are

$$V_h(q) = \frac{r + 6c - a}{2}|q|^2 + \frac{g}{4} \sum_e q_e^4 + \frac{\lambda}{4} \sum_{\{e,f\} \in E(Q_3)} (q_e - q_f)^2 (q_e^2 + q_f^2) - h(u, q), \quad (5.3)$$

$$J_{yz} = c \mathbf{1}_{\{|y-z|=1\}}, \quad J_0 = 6c. \quad (5.4)$$

For a finite Δ and boundary loop ξ , the specification is

$$\begin{aligned} I_\Delta^h(\omega \mid \xi) &= \sum_{y \in \Delta} \int_0^\beta V_h(\omega_y) d\tau - \frac{1}{2} \sum_{y,z \in \Delta} J_{yz}(\omega_y, \omega_z)_{L^2} \\ &\quad - \sum_{y \in \Delta, z \notin \Delta} J_{yz}(\omega_y, \xi_z)_{L^2}, \end{aligned} \quad (5.5)$$

$$\pi_\Delta^h(f \mid \xi) = Z_\Delta(h, \xi)^{-1} \int f(\omega_\Delta \xi_{\Delta^c}) e^{-I_\Delta^h(\omega \mid \xi)} \prod_{y \in \Delta} \chi_a(d\omega_y). \quad (5.6)$$

Only the negative pair interaction accompanies V_h ; a second positive spatial diagonal would double-count the bond energy.

Fix a source window $|h| \leq h_0$ and set

$$\begin{aligned} b &= \tfrac{1}{2}|r + 6c - a|, & A &= g/128, \\ C_0 &= 16b^2/g + \tfrac{3}{4}h_0^{4/3}(32/g)^{1/3}, \\ V^+(q) &= (g/4 + 3\lambda) \sum_e q_e^4 + b|q|^2 + h_0|q|. \end{aligned} \quad (5.7)$$

Then, uniformly on this window,

$$A|q|^4 - C_0 \leq V_h(q) \leq V^+(q). \quad (5.8)$$

Indeed, $\sum_e q_e^4 \geq |q|^4/8$; absorb $b|q|^2$ against $g|q|^4/64$ and $h_0|q|$ against $g|q|^4/128$. The respective scalar maxima are the two terms in C_0 , leaving $A|q|^4$. For the upper bound, $(x - y)^2(x^2 + y^2) \leq 4(x^4 + y^4)$ and the degree of Q_3 is three. All envelopes are continuous and $V_h(0) = V^+(0) = 0$.

Here is the precise general-vector use of Kozitsky–Pasurek (KP) [2]. In their Assumption (A), (2.5), take dimension $\nu = 8$, growth exponent $r_{\text{KP}} = 2$, lower constants $A_V = A$, $B_V = -C_0$, and upper envelope V^+ . Positive m, a specify the oscillator reference. Their interaction condition (2.6) holds with symmetric zero-diagonal J and row sum $J_0 = 6c$; the onsite locking is not part of J . Lattice regularity (2.1) follows from $\sum_{y \in \mathbb{Z}^3} (1 + |y|)^{-3-\varepsilon} < \infty$ for every $\varepsilon > 0$. For Assumption (B), use the exponential weights of (2.42)–(2.43): $J_\alpha = 6ce^\alpha < \infty$, $J_\alpha - J_0 \rightarrow 0$, and the logarithmically weighted sum in (2.39) is finite. Their Theorem 3.1 therefore supplies a nonempty W_t -compact set $\mathcal{G}_t(h)$ at each fixed source. Theorem 3.2 gives a moment bound for this fixed-model set, not automatically for a source window or finite tori. The topology agrees with KP (2.47)–(2.49): both require all weighted L^2 norms and all local sup norms. Replacing positive summable local weights or $s/(1+s)$ by $1 \wedge s$ leaves that topology unchanged; $\alpha_k \downarrow 0$ gives a countable cofinal family of weighted norms. No scalar or rotation-invariant phase theorem is used. We prove the needed periodic and varying-source statements directly below.

5.2 A source-window bound uniform in periodic volume

For the oscillator Gaussian choose $\ell_\sigma > 0$ so that $F_\sigma := \int e^{\ell_\sigma \|\omega\|_{C^\sigma}^2} \chi_a(d\omega) < \infty$ (KP Proposition 2.2, (2.27)). Fix $\kappa, \theta > 0$ and define

$$\begin{aligned} Y_- &= \int \exp \left[-\frac{J_0}{2\theta} \|\omega\|_{L^2}^2 - \int_0^\beta V^+(\omega) d\tau \right] \chi_a(d\omega), \\ D &= \kappa + J_0/(2\theta), & C_1 &= \max\{0, \log F_\sigma + \beta C_0 + \beta D^2/(4A) - \log Y_-\}, \\ f_y(\omega) &= \ell_\sigma \|\omega_y\|_{C^\sigma}^2 + \kappa \|\omega_y\|_{L^2}^2. \end{aligned} \quad (5.9)$$

The integrand defining Y_- is positive and at most one, so $0 < Y_- \leq 1$. Bilinear Young bounds give, for every neighbor boundary,

$$Z_{\{y\}}(h, \xi) \geq Y_- \exp \left[-\frac{\theta}{2} \sum_z J_{yz} \|\xi_z\|_{L^2}^2 \right],$$

$$\int e^{f_y - I_{\{y\}}^h} d\chi_a \leq F_\sigma e^{\beta C_0 + \beta D^2/(4A)} \exp \left[\frac{\theta}{2} \sum_z J_{yz} \|\xi_z\|_{L^2}^2 \right].$$

The second inequality uses $Ds^2 - As^4 \leq D^2/(4A)$ pointwise in time. Division, with the positive lower bound already established, yields

$$\pi_{\{y\}}^h(e^{f_y} \mid \xi) \leq \exp \left[C_1 + \theta \sum_z J_{yz} \|\xi_z\|_{L^2}^2 \right]. \quad (5.10)$$

This is the finite-range, source-window version of the mechanism in KP Lemma 4.1. Its constants do not depend on the site or torus size.

Let $L \geq 4$ be even. Under the periodic law, translation symmetry makes $M = \mathbb{E}_{L,h} e^{f_y}$ independent of y . First $M < \infty$: in the finite product Gaussian integral, (5.8) and $|\sum_{\{y,z\}} c(\omega_y, \omega_z)_{L^2}| \leq (J_0/2) \sum_y \|\omega_y\|_{L^2}^2$ leave a positive quartic at every site. It absorbs all quadratic terms, including $\kappa \|\omega_y\|_{L^2}^2$; the remaining $e^{\ell_\sigma \|\omega_y\|_{C^\sigma}^2}$ is Gaussian integrable. The finite normalization is positive. This proves finiteness before any use of a power of M .

Set $t = \theta J_0/\kappa < 1$. Conditional expectation, (5.10), generalized Holder with weights J_{yz}/J_0 , and concavity of x^t imply

$$M \leq e^{C_1} \mathbb{E}_{L,h} \exp \left(\theta \sum_z J_{yz} \|\omega_z\|_{L^2}^2 \right)$$

$$\leq e^{C_1} \prod_{z: J_{yz} > 0} (\mathbb{E}_{L,h} e^{t f_z})^{J_{yz}/J_0} = e^{C_1} \mathbb{E}_{L,h} e^{t f_y} \leq e^{C_1} M^t. \quad (5.11)$$

Choose $\theta = \kappa/(2J_0)$ and evaluate C_1 at this choice. Since M is finite and positive, $M \leq e^{2C_1} =: C_{\text{per}}$. In particular

$$\sup_{L \geq 4, |h| \leq h_0, y} \mathbb{E}_{L,h} \exp \left(\ell_\sigma \|\omega_y\|_{C^\sigma}^2 + \kappa \|\omega_y\|_{L^2}^2 \right) \leq C_{\text{per}}. \quad (5.12)$$

Extend the law by zero outside the centered cube $[-L/2, L/2]^3 \cap \mathbb{Z}^3$, denoting it by $\hat{\mu}_{L,h}$. Since $C_{\text{per}} \geq 1$, the estimate holds also at exterior sites. In particular $\hat{\mu}_{L,h}(\|\omega_y\|_{C^\sigma}^2) \leq C_{\text{per}}/\ell_\sigma$.

5.3 Weighted compactness with the correct tail direction

Put $\|\omega\|_{\alpha, \sigma}^2 = \sum_y e^{-\alpha|y|} \|\omega_y\|_{C^\sigma}^2$ and $S(\alpha) = \sum_y e^{-\alpha|y|} < \infty$. For a given $\varepsilon > 0$, choose positive ε_k with $\sum_k \varepsilon_k \leq \varepsilon$, and radii satisfying $R_k^2 \geq C_{\text{per}} S(\alpha_k)/(\ell_\sigma \varepsilon_k)$. Markov's inequality and a union bound show that

$$K_\varepsilon = \{\omega \in \Omega_t : \|\omega\|_{\alpha_k, \sigma} \leq R_k \text{ for all } k\}, \quad \inf_{L,h} \hat{\mu}_{L,h}(K_\varepsilon) \geq 1 - \varepsilon. \quad (5.13)$$

This set is compact in d_t . To see this, its bounds give equibounded and equicontinuous loops at each site, hence a diagonal locally uniform subsequence by Arzela–Ascoli. For the weighted L^2 norm at α_k , use the *stronger* bound at $\alpha_{k+1} < \alpha_k$:

$$\sum_{|y| > R} e^{-\alpha_k|y|} \|\omega_y\|_{L^2}^2 \leq \beta e^{-(\alpha_k - \alpha_{k+1})R} R_{k+1}^2. \quad (5.14)$$

This tends to zero uniformly on K_ε . Finite-site convergence therefore implies convergence in every weighted norm (apply the same tail bound to both terms of a difference). Lower semicontinuity of the Holder norm under uniform convergence and Fatou's lemma put the limit back in K_ε . This proves compactness and, by Prokhorov, common W_t tightness on the entire source window. The reverse weight direction does not work: at $\beta = 1$ a constant loop with squared amplitude 4^n at site $(n, 0, 0)$ has squared $\log 4$ -weighted norm one but squared $\log 2$ -weighted norm 2^n . No spatial tightness is inferred from a fixed-volume Gaussian bound.

5.4 Positive finite normalizers and continuous DLR kernels

For a nonempty finite Δ of size n and a compact $K \subset \Omega_t$, write

$$\begin{aligned} Q_\Delta(\omega) &= \sum_{y \in \Delta} \int_0^\beta |\omega_y|^4 d\tau, \\ B_K &= \frac{1}{2} \sup_{\xi \in K} \sum_{y \in \Delta, z \notin \Delta} J_{yz} \|\xi_z\|_{L^2}^2, \\ C_{\Delta,K} &= \beta n(C_0 + J_0^2/(8A)) + B_K. \end{aligned}$$

Only finitely many boundary sites enter, so $B_K < \infty$. The internal and exterior bilinear Young bounds allocate in total at most $J_0/2$ to the quadratic norm of each interior loop. Consequently

$$\begin{aligned} I_\Delta^h &\geq A Q_\Delta - \beta n C_0 - \frac{J_0}{2} \sum_{y \in \Delta} \|\omega_y\|_{L^2}^2 - B_K \\ &\geq \frac{A}{2} Q_\Delta - C_{\Delta,K}, \quad |h| \leq h_0, \quad \xi \in K, \end{aligned} \quad (5.15)$$

where the time-volume factor is tracked as follows. Put $S_y = \|\omega_y\|_{L^2}^2$. Cauchy-Schwarz gives $S_y^2 \leq \beta \int_0^\beta |\omega_y(\tau)|^4 d\tau$. Hence

$$\frac{A}{2} \int_0^\beta |\omega_y|^4 d\tau + \frac{\beta J_0^2}{8A} \geq \frac{A}{2\beta} S_y^2 + \frac{\beta J_0^2}{8A} \geq \frac{J_0}{2} S_y,$$

which, after summing over y , gives the displayed $C_{\Delta,K} = \beta n(C_0 + J_0^2/(8A)) + B_K$. This explicit normalization is the only place where the imaginary-time length enters this coercivity estimate.

For a noncircular lower bound on Z_Δ , set $R_F^2 = \log(2F_\sigma)/\ell_\sigma$. Gaussian Markov gives $\chi_a(\|\omega\|_\infty \leq R_F) \geq 1/2$. Define

$$\begin{aligned} T_K &= \sup_{\xi \in K} \sum_{y \in \Delta, z \notin \Delta} J_{yz} \|\xi_z\|_{L^2}, \\ U_{\Delta,K} &= \beta n\{(g/4 + 3\lambda)R_F^4 + (b + J_0/2)R_F^2 + h_0 R_F\} + \sqrt{\beta} R_F T_K. \end{aligned}$$

On the product event $\|\omega_y\|_\infty \leq R_F$ at every interior site, $I_\Delta^h \leq U_{\Delta,K}$, by (5.8) and Cauchy-Schwarz on the boundary terms. This event has Gaussian probability at least 2^{-n} . Hence

$$\inf_{|h| \leq h_0, \xi \in K} Z_\Delta(h, \xi) \geq z_{\Delta,K} := 2^{-n} e^{-U_{\Delta,K}} > 0. \quad (5.16)$$

Finiteness follows separately from (5.15).

The same bounds give a useful quantitative source estimate. Let $X_\Delta = \sum_{y \in \Delta} \int_0^\beta (u, \omega_y) d\tau$. Holder's inequality gives $|X_\Delta| \leq (\beta n)^{3/4} Q_\Delta^{1/4}$. Because $I_\Delta^h = I_\Delta^0 - h X_\Delta$, the mean value theorem and (5.15) bound the difference of two unnormalized weights by

$$|h - h'| e^{C_{\Delta,K}} (\beta n)^{3/4} Q_\Delta^{1/4} e^{-A Q_\Delta/2} \leq |h - h'| M_{\Delta,K}, \quad M_{\Delta,K} = e^{C_{\Delta,K}} (\beta n)^{3/4} (2Ae)^{-1/4}.$$

The scalar maximum occurs at $Q_\Delta = 1/(2A)$. The numerator and normalizer thus have respective Lipschitz bounds $\|f\|_\infty M_{\Delta,K}$ and $M_{\Delta,K}$. Since the absolute numerator is at most $\|f\|_\infty$ times its own normalizer, division gives, for every bounded Borel f ,

$$\sup_{\xi \in K} |\pi_\Delta^h(f | \xi) - \pi_\Delta^{h'}(f | \xi)| \leq \frac{2M_{\Delta,K}}{z_{\Delta,K}} \|f\|_\infty |h - h'|. \quad (5.17)$$

If $f \in C_b(\Omega_t)$ and $\xi_j \rightarrow \xi$ in d_t , replacing finitely many interior loops is continuous, and the finite-range boundary pairings converge in L^2 . The compact set consisting of this sequence and its limit admits (5.15) and (5.16). Dominated convergence of the numerator and denominator proves $\pi_\Delta^h f \in C_b(\Omega_t)$. The kernels replace only finitely many loops, preserve Ω_t , and satisfy specification consistency by conditional integration and Fubini. This direct finite-range Feller proof avoids assuming the desired DLR limit in order to bound its kernel. Since bounded continuous functions on the Polish space determine probability measures, the equalities $\mu(f) = \mu(\pi_\Delta^h f)$ for all such f imply the full Borel DLR equation $\mu \pi_\Delta^h = \mu$.

5.5 Periodic limits and the two source-tangent passages

Fix h . Once Δ and all its neighbors lie strictly within the centered cube, the conditional law of $\hat{\mu}_{L,h}$ on Δ equals (5.6); no periodic seam meets an incident bond. Thus $\hat{\mu}_{L,h}(f) = \hat{\mu}_{L,h}(\pi_\Delta^h f)$. Common tightness and the Feller property pass this identity along every converging periodic subsequence, first for $C_b(\Omega_t)$ and then for Borel sets as above. Each accumulation point is therefore in $\mathcal{G}_t(h)$. For every fixed local continuous cylinder test, its support and any fixed translate eventually lie within the cube; torus symmetry then passes to the limit. Such tests determine the product Borel law, proving spatial translation invariance of the limit. The finite zero extension itself is not spatially translation invariant. Time-shift invariance passes too; time shifts are continuous in d_t . By lower semicontinuity and truncation, all selected limits inherit (5.12).

At a differentiability point h of P_β , let μ_h be any such accumulation point. With $Q_0(\omega) = (u, \omega_0(0))$, define $\text{clip}_R(s) = \max(-R, \min(s, R))$. For the periodic laws and their selected limits,

$$\mathbb{E}Q_0^2 \leq C_{\text{per}}/\ell_\sigma, \quad \mathbb{E}|Q_0 - \text{clip}_R(Q_0)| \leq C_{\text{per}}/(\ell_\sigma R). \quad (5.18)$$

Abbreviate $p_L = p_{\beta,L}$ and $P_L = p_L/(8\beta)$ at this fixed β . For finite L , spatial and time translation symmetry and the actual source e^{hX_L} give $p'_L(h) = \mathbb{E}X_L/V = \beta\mathbb{E}Q_0$ and $P'_L(h) = \mathbb{E}Q_0/8$. Section 4 gives $P'_L(h) \rightarrow P'_\beta(h)$ at differentiability points. Pass bounded clips through weak convergence and then let $R \rightarrow \infty$ in (5.18); this proves

$$\mu_h(Q_0) = 8P'_\beta(h). \quad (5.19)$$

There is no extra factor β in the normalized tangent identity.

Choose positive differentiability points $h_j \downarrow 0$. A finite convex pressure has at most countably many nondifferentiability points. Monotonicity and the secant inequalities imply $P'_\beta(h_j) \rightarrow D_+P_\beta(0)$. More explicitly, $P'_\beta(h_j) \geq D_+P_\beta(0)$; for fixed $t > 0$ and $h_j < t$, convexity gives $P'_\beta(h_j) \leq [P_\beta(t) - P_\beta(h_j)]/(t - h_j)$. Take $j \rightarrow \infty$, then $t \downarrow 0$. The endpoint derivative is finite because zero is interior to the finite-pressure domain. The common source-window bound gives a subsequence $\mu_{h_j} \Rightarrow \mu_+$ in W_t . For $f \in C_b(\Omega_t)$, split the difference between the h_j and zero kernels over K_ε and its complement. Equations (5.13) and (5.17) give

$$|\mu_{h_j}(\pi_\Delta^{h_j} f - \pi_\Delta^0 f)| \leq \frac{2M_{\Delta,K_\varepsilon}}{z_{\Delta,K_\varepsilon}} \|f\|_\infty h_j + 2\|f\|_\infty \varepsilon.$$

The compact-set probability bound holds for these limits because they inherit the moment estimate. For fixed ε , take $j \rightarrow \infty$; the zero-source Feller property passes the remaining identity. Then let $\varepsilon \downarrow 0$. This proves the full zero-source DLR equation. Finally apply (5.18) once more, now uniformly in j , to pass the unbounded local expectation. Together with (5.19) this gives

$$\mu_+ \in \mathcal{G}_t(0), \quad \mu_+(Q_0) = 8D_+P_\beta(0), \quad (5.20)$$

without inferring unbounded-moment convergence from weak convergence alone. The global parity map $\Theta\omega = -\omega$ preserves Ω_t and intertwines the zero-source specification with itself. Hence $\Theta_*\mu_+ \in \mathcal{G}_t(0)$ with opposite collective expectation. Distinctness still requires the strictly positive cusp established later; it does not follow merely from this construction. Neither uniqueness nor exhaustion of all tempered DLR states by periodic limits is asserted. The order is the fixed-volume loop limit, spatial accumulation at each fixed h , then $h_j \downarrow 0$; no arbitrary simultaneous $h = h(L)$ or $\beta \rightarrow \infty$ passage is used.

6 Continuous-loop FKG and collective covariance

Order loops coordinatewise in space, component, and imaginary time. A law is associated if $\text{Cov}(F, G) \geq 0$ for bounded increasing Borel functions F, G . At mesh N , the density is $\rho_N \propto e^{-S_N}$, where the temporal quadratic bonds, spatial bonds, and internal Q_3 terms satisfy

$$\partial_i \partial_j \log \rho_N = -\partial_i \partial_j S_N \geq 0 \quad (i \neq j), \quad -\partial_x \partial_y \frac{\lambda}{4} (x-y)^2 (x^2 + y^2) = \frac{\lambda}{4} [(x+y)^2 + 5(x-y)^2] \geq 0. \quad (6.1)$$

The first inequality concerns the log density, with the opposite sign to the energy Hessian. Temporal and spatial bonds contribute m/ε and εc per occurrence, respectively, and the internal contribution is ε times the displayed polynomial. Unary terms contribute zero.

Here is the finite-dimensional association argument underlying the FKG mechanism [4]. First restrict a positive smooth density ρ with $\partial_i \partial_j \log \rho \geq 0$ to a compact product interval. In one dimension, for independent copies T, T' ,

$$\text{Cov}(f(T), g(T)) = \frac{1}{2} \mathbb{E}[(f(T) - f(T'))(g(T) - g(T'))] \geq 0$$

for bounded increasing f, g . Induct on dimension, conditioning on the last coordinate t . The conditional laws satisfy the induction hypothesis. For $t_2 > t_1$ their unnormalized likelihood ratio $R(x) = \rho(x, t_2)/\rho(x, t_1)$ is bounded and increasing, because

$$\partial_i \log R(x) = \int_{t_1}^{t_2} \partial_i \partial_t \log \rho(x, t) dt \geq 0.$$

Conditional association implies $\mathbb{E}_{t_1}(fR) \geq \mathbb{E}_{t_1} f \mathbb{E}_{t_1} R$, hence $\mathbb{E}_{t_2} f \geq \mathbb{E}_{t_1} f$. For increasing $F(x, t)$, increase t inside F and then increase the conditional law; this shows that $m_F(t) = \mathbb{E}_t F(x, t)$ is increasing. The same applies to G , so

$$\text{Cov}(F, G) = \mathbb{E} \text{Cov}(F, G | t) + \text{Cov}(m_F(t), m_G(t)) \geq 0.$$

Apply the result to ρ_N on growing product cubes. Its integrability and dominated convergence for the three bounded expectations remove the cutoff, proving association on the full mesh space.

6.1 Passage to bounded continuous and Borel loop tests

Periodic linear interpolation is order preserving. For bounded continuous increasing F, G on $\mathbf{E}_L$, finite association therefore applies to $F \circ \mathcal{I}_N$ and $G \circ \mathcal{I}_N$. Apply the actual interacting weak convergence (3.15) to the three bounded continuous functions F, G, FG . Their limiting covariance is nonnegative; FG need not be increasing for this weak-convergence step.

Here is the measurable extension used below. We first verify the topological hypotheses for the actual tempered space rather than leave them implicit.

Lemma 6.1 (Positive cone in the tempered topology). *With the metric d_t of (5.2), the cone $\Omega_{t,+} = \{\omega \in \Omega_t : \omega_y(\tau) \geq 0 \text{ for all } y, \tau\}$ is closed and d_t is translation invariant. If $K \subset \Omega_t$ is compact, then $K + \Omega_{t,+}$ is closed and is an upper set.*

Proof. If $\omega^{(n)} \rightarrow \omega$ in d_t , each local sup-norm term in (5.2) tends to zero, so $\omega_y^{(n)} \rightarrow \omega_y$ uniformly for every fixed site y . Nonnegativity is therefore preserved in the limit, proving that $\Omega_{t,+}$ is closed. Every summand in d_t depends only on a difference, so $d_t(\omega + a, \eta + a) = d_t(\omega, \eta)$. For closedness of $K + \Omega_{t,+}$, take $k_n + e_n \rightarrow x$ with $k_n \in K$ and $e_n \in \Omega_{t,+}$. After passing to a subsequence, $k_n \rightarrow k \in K$. Translation invariance and the triangle inequality give

$$d_t(e_n, x - k) \leq d_t(k_n + e_n, x) + d_t(k_n + x - k, x) \rightarrow 0,$$

so $e_n \rightarrow x - k$. The cone is closed, hence $x - k \in \Omega_{t,+}$ and $x = k + (x - k) \in K + \Omega_{t,+}$. Finally, adding any $v \in \Omega_{t,+}$ replaces e by $e + v$, so the sum is upper. $\square$

With $E_+ = \Omega_{t,+}$, the following measurable extension applies in a Polish topological vector space with a closed positive cone and a compatible translation-invariant metric d . Let C be a closed upper set. If $x \leq x'$, then $z + (x' - x) \in C$ for $z \in C$ and $d(x', z + (x' - x)) = d(x, z)$. Thus $d(x, C)$ is nonincreasing in x . The bounded continuous increasing functions $f_j(x) = \max(0, 1 - jd(x, C))$ decrease to 1_C ; use $f_j = 0$ if C is empty. Bounded convergence transfers positive correlation to pairs of closed upper sets.

For upper Borel sets A, B , choose compact subsets $K \subset A$ and $J \subset B$ whose probabilities approximate theirs from below. The sets $K + E_+$ and $J + E_+$ are closed upper sets contained in A, B :

compactness allows extraction of the compact summand, and closedness of the cone then passes the other summand to the limit. Hence

$$\mu(A \cap B) \geq \mu((K + E_+) \cap (J + E_+)) \geq \mu(K + E_+)\mu(J + E_+) \geq \mu(K)\mu(J).$$

Inner regularity proves positive correlation for A, B . Integrating this inequality over upper level sets proves association for nonnegative bounded increasing Borel functions. Adding constants handles signed functions. The finite loop space has all the stated properties with its sup metric and pointwise positive cone. This proves continuous-loop association at every fixed L, h .

For the selected spatial limits of Section 5, zero extension is order preserving. Their weak convergence in the declared weighted topology therefore passes association for bounded continuous tests. The same argument applies along the subsequent source-tangent sequence; the Borel extension uses the closed cone and compatible translation-invariant metric of that Polish projective space. It does not show association for arbitrary mixtures of DLR states. For example, the equally weighted mixture of point masses at $(0, 1)$ and $(1, 0)$ has coordinate covariance $-1/4$, although each point mass is associated.

6.2 Coordinate products and the collective inequalities

At fixed volume, (3.19) permits clipping of coordinate products. Along the spatial/source sequences, (5.12) instead gives a common bound $\mathbb{E}|\omega_{y,e}(t)|^4 \leq C_4 := 2C_{\text{per}}/\ell_\sigma^2$, by $z^2 \leq 2e^z$ with $z = \ell_\sigma \|\omega_y\|_{C^\sigma}^2$. For two coordinates Y, Z and $\text{clip}_R(t) = \max(-R, \min(t, R))$, association applies to their increasing clips. Splitting the error over $\{|Y| > R\}$ and $\{|Z| > R\}$, Cauchy–Schwarz and the fourth-moment Markov bound give

$$\mathbb{E}|YZ - \text{clip}_R(Y) \text{clip}_R(Z)| \leq 2C_4/R^2. \quad (6.2)$$

Clipped first moments converge as well, for example with error at most C_4/R^3 . This both removes the clips and permits passage of the expectations through the stated limits.

For the finite periodic zero-source law and its parity-invariant periodic limits, the coordinate means vanish. This parity statement is not imposed on symmetry-breaking source-tangent states at zero source. Consequently, on the finite zero-source law, for distinct coordinates,

$$\mathbb{E}q_{y,e}(\tau)q_{z,f}(s) \geq 0. \quad (6.3)$$

At one site, the three-regularity of Q_3 gives

$$\mathbb{E}D_0^{\text{int}} = 3\mathbb{E}S_0 - 2 \sum_{\{e,f\} \in E(Q_3)} \mathbb{E}(q_{0,e}q_{0,f}) \leq 3\mathbb{E}S_0, \quad (6.4)$$

$$\mathbb{E}Q_0^2 = \frac{1}{8} \left(\mathbb{E}S_0 + 2 \sum_{e < f} \mathbb{E}(q_{0,e}q_{0,f}) \right) \geq \frac{1}{8} \mathbb{E}S_0. \quad (6.5)$$

These are expectation inequalities, not pointwise inequalities for arbitrary vectors.

7 Reflection positivity and the infrared bound

7.1 A bounded positive kernel on the loop Hilbert space

Write $B_h(q)$ for the physical onsite potential in (2.2), excluding spatial bonds. With the normalized oscillator Gaussian χ_a of Section 5, define a local finite measure

$$\nu_h(d\omega) = \exp \left[- \int_0^\beta \{B_h(\omega(\tau)) - a|\omega(\tau)|^2/2\} d\tau \right] \chi_a(d\omega). \quad (7.1)$$

Its density is strictly positive and bounded above by a finite constant: the scalar quartic bounds the residual polynomial below, and the locking term is nonnegative. This uses the positive-difference spatial convention; there is no additional allocated $3c|q|^2$ in this local measure. The exact periodic law

is proportional to $\prod_y \nu_h(d\omega_y) \prod_{\{y,z\} \in \mathcal{E}_L} K(\omega_y, \omega_z)$, where $\mathcal{E}_L$ consists of the nearest-neighbor bonds counted once and

$$K(v, w) = \exp[-(c/2) \|v - w\|_{\mathbf{H}}^2], \quad \mathbf{H} = L^2([0, \beta]; \mathbb{R}^8). \quad (7.2)$$

Because $0 < K \leq 1$, the partition integral is finite and positive. No infinite-dimensional Lebesgue measure is being used.

To prove positive definiteness, take finite-rank orthogonal projections $P_n \rightarrow 1$ strongly on $\mathbf{H}$. On their ranges, of dimension d_n , the ordinary Gaussian Fourier formula is

$$K_n(v, w) := e^{-c|P_n(v-w)|^2/2} = \int_{\mathbb{R}^{d_n}} e^{ik \cdot P_n v} \overline{e^{ik \cdot P_n w}} \frac{e^{-|k|^2/(2c)}}{(2\pi c)^{d_n/2}} dk.$$

For every finite complex measure ζ on $\mathbf{H}$, Fubini therefore gives

$$\iint K_n(v, w) \zeta(dv) \overline{\zeta(dw)} \geq 0. \quad (7.3)$$

Since $|K_n| \leq 1$ and $K_n \rightarrow K$ pointwise, dominated convergence for $|\zeta| \otimes |\zeta|$ gives the same inequality for K . The argument also applies on $\mathbf{H}^M$ for any finite number M of crossing bonds. It does not postulate a Gaussian with identity covariance on an infinite Hilbert space or require a positive Fourier transform of the onsite quartic.

7.2 Direct spatial reflection positivity

For even $L \geq 4$, use representatives $0, \dots, L-1$ and reflect by $\vartheta(y_1, y_2, y_3) = (L-1-y_1, y_2, y_3)$. Let $\Lambda_+ = \{y : y_1 < L/2\}$ and $\Lambda_- = \vartheta\Lambda_+$. The two periodic separating planes lie between sites. Each crossing bond joins $y \in \Lambda_+$ to ϑy , and there are $M = 2L^2$ such bonds. Let $b(v)$ list their plus-side loops in $\mathbf{H}^M$. The finite measure α_h on plus-half configurations is the product of their local ν_h measures times all bonds internal to that half. Reflection identifies the minus half with a second copy of the same measure. The full unnormalized law is consequently

$$\alpha_h(dv) \alpha_h(dw) e^{-c\|b(v)-b(w)\|_{\mathbf{H}^M}^2/2}.$$

For a bounded complex Borel function F of the plus half, push the finite complex measure $F(v)\alpha_h(dv)$ through b and use (7.3). After division by the positive normalizer, this proves

$$\mathbb{E}_{L,h}[F\Theta_\vartheta F] \geq 0, \quad \Theta_\vartheta F(\omega) = \overline{F(\vartheta\omega)}. \quad (7.4)$$

The other coordinate directions are identical. This includes bounded measurable tests and any spatially constant real source h . It is spatial reflection positivity, not a real-time reconstruction statement. Gaussian domination is invoked through the finite-dimensional theorem below, not inferred from reflection positivity in a single unexplained step.

7.3 The finite-dimensional FSS input and its exact source map

The external input is Froehlich–Simon–Spencer (FSS) [3], Section 2, Theorem 2.1, printed page 81, with proof on pages 82–84. In the notation used here, it applies to finite-dimensional vector spins on a periodic rectangular nearest-neighbor lattice, with density proportional to $e^{J \sum_{\{y,z\}} s_y \cdot s_z} \prod_y \varpi(ds_y)$, $J > 0$. The common finite prior must satisfy $\int e^{a_1 |s|^2} \varpi(ds) < \infty$ for every $a_1 > 0$; radiality is not assumed. For an edge field $\mathfrak{h}$ and its adjoint-divergence source $B\mathfrak{h}$, the theorem is

$$\mathbb{E} \exp \left(\sum_y s_y \cdot (B\mathfrak{h})_y \right) \leq \exp \left(\frac{\|\mathfrak{h}\|^2}{2J} \right). \quad (7.5)$$

This is a finite-dimensional input, not an imported loop theorem. Its constant has no dependence on spin dimension or prior moment constants. The primary-source recheck recorded in EXP-001652

confirms this exact fixed-mesh scope and leaves the mesh-to-loop, thermodynamic, DLR, cusp, and parity-state passages as Q3LOCK-local obligations.

At a fixed time mesh $\varepsilon = \beta/N$, set $s_y = (\sqrt{\varepsilon}x_{y,k})_{k=0}^{N-1} \in \mathbb{R}^{8N}$. The actual zero-source action, with the auxiliary harmonic split canceled, is

$$\begin{aligned} S_{N,L,0}(x) &= \sum_y \frac{m}{2\varepsilon} \sum_k |x_{y,k+1} - x_{y,k}|^2 + \varepsilon \sum_{y,k} B_0(x_{y,k}) + \frac{c\varepsilon}{2} \sum_{\{y,z\},k} |x_{y,k} - x_{z,k}|^2 \\ &= \sum_y \mathcal{V}_N(s_y) - c \sum_{\{y,z\}} s_y \cdot s_z, \end{aligned} \quad (7.6)$$

where $\mathcal{V}_N$ includes the temporal term and $\varepsilon \sum_k \{B_0(x_k) + 3c|x_k|^2\}$. Thus $J = c$ and the same prior $e^{-\mathcal{V}_N(s)} ds$ is used at every spatial site. Its normalization cancels in Gibbs expectations. Nonnegative temporal and locking terms and Cauchy–Schwarz on the $8N$ coordinates give

$$\mathcal{V}_N(s) \geq \frac{g}{4\varepsilon} \sum_{k,e} s_{k,e}^4 + \frac{r+6c}{2} |s|^2 \geq \frac{g}{32\beta} |s|^4 + \frac{r+6c}{2} |s|^2. \quad (7.7)$$

This proves finiteness and all quadratic exponential moments at every fixed N . The dimension and prior integrals may depend on N ; we need only the dimension-free source constant in (7.5).

Here are explicit operator domains and signs. On vertex fields define $(Gv)(y, j) = v_{y-e_j} - v_y$, and on edge fields define $(Ba)_y = \sum_j \{a(y + e_j, j) - a(y, j)\}$. Summation by parts gives $B = G^*$ and $L_{\text{sp}} = G^*G = BB^*$. This realizes the forward differences in the FSS divergence and their backward adjoints. The operators act componentwise on $\mathbb{R}^{8N}$. Restrict L_{sp}^{-1} to the zero-sum vertex space V_0 ; its kernel on the full space is the constants. For a real spatial field f with $\sum_y f_y = 0$, define

$$X_L(f) = \int_0^\beta \sum_y f_y Q_y(\tau) d\tau, \quad K_L(f) = \frac{\beta}{2c} \langle f, L_{\text{sp}}^{-1} f \rangle. \quad (7.8)$$

For real t , set

$$\begin{aligned} \eta_y(t) &= t\sqrt{\varepsilon}(f_y u)_{k=0}^{N-1}, \quad \mathfrak{h}_t = GL_{\text{sp}}^{-1}\eta(t), \\ B\mathfrak{h}_t &= \eta(t), \quad \|\mathfrak{h}_t\|^2 = \langle \eta(t), L_{\text{sp}}^{-1}\eta(t) \rangle = \beta t^2 \langle f, L_{\text{sp}}^{-1} f \rangle. \end{aligned} \quad (7.9)$$

The last equality uses $N\varepsilon = \beta$ and $|u| = 1$; there is no factor eight. The source pairing is exactly $\sum_y s_y \cdot \eta_y(t) = tX_{N,L}(f)$, where $X_{N,L}(f) = \varepsilon \sum_{y,k} f_y(u, x_{y,k})$. The vertex norm $\|\eta\|^2$ cannot replace the edge Poisson norm. Equivalently, the spatial square completion is

$$\frac{c}{2} \|Gs\|^2 - \langle \eta, s \rangle = \frac{c}{2} \|Gs - \mathfrak{h}_t/c\|^2 - \frac{1}{2c} \|\mathfrak{h}_t\|^2. \quad (7.10)$$

Its bond displacement is $\mathfrak{h}_t/c$, not $\mathfrak{h}_t$. Applying (7.5) to the exact prior yields

$$\mathbb{E}_{N,L,0} e^{tX_{N,L}(f)} \leq e^{K_L(f)t^2} \quad (t \in \mathbb{R}). \quad (7.11)$$

7.4 Actual loop passage and source uniform integrability

Periodic polygonal interpolation satisfies the exact cyclic integral identity $X_L(f)(\mathcal{I}_N x) = X_{N,L}(f)$. The functional $X_L(f)$ is continuous for the finite-volume loop sup norm. The interacting weak limit (3.15) therefore gives convergence in distribution of these scalar variables. Applying it to $\min(e^{tX_L(f)}, R)$ and then taking $R \uparrow \infty$ transfers (7.11) without first assuming convergence of unbounded expectations:

$$\mathbb{E}_{L,0} \exp(tX_L(f)) \leq \exp(K_L(f)t^2). \quad (7.12)$$

In fact the MGFs and the required derivatives converge. For $T \geq 0$, the two signs of (7.11) give

$$\sup_N \mathbb{E}_{N,L,0} e^{T|X_{N,L}(f)|} \leq 2e^{K_L(f)T^2}. \quad (7.13)$$

Using $T = 2|t|$ bounds the squared L^2 norm of $e^{tX_{N,L}(f)}$, which proves uniform integrability. More generally, for every integer $k \geq 0$ and auxiliary $a_2 > 0$, the exponential series implies

$$|x|^{2k} e^{2T|x|} \leq (2k)! a_2^{-2k} e^{(2T+a_2)|x|}.$$

Together with (7.13) this bounds the L^2 norms of $|X_{N,L}|^k e^{T|X_{N,L}|}$ uniformly in N . Truncation and weak convergence now pass these source moments, in particular $\mathbb{E}X_{N,L}^2 \rightarrow \mathbb{E}X_L^2$. The parameter a_2 has reciprocal source- observable units and introduces no physical assumption. Means vanish at zero source by parity. All these estimates are at fixed L, β, f ; they neither control the spatial constant mode nor replace the general-source quartic bounds or the volume-uniform DLR argument.

7.5 Duhamel normalization and the Fourier eigenvalue

Define the time-averaged Duhamel matrix

$$C_L(y, z; \tau) = \mathbb{E}[Q_y(\tau)Q_z(0)], \quad D_L(y, z) = \frac{1}{\beta} \int_0^\beta C_L(y, z; \tau) d\tau. \quad (7.14)$$

All expectations here are in the exact zero-source periodic loop law. Its finite coordinate moments justify Fubini. With $\overline{Q}_y = \beta^{-1} \int_0^\beta Q_y(\tau) d\tau$, time-shift invariance and parity give

$$\mathbb{E}\overline{Q}_y\overline{Q}_z = \frac{1}{\beta^2} \int_0^\beta \int_0^\beta C_L(y, z; \tau - \sigma) d\tau d\sigma = D_L(y, z), \quad \text{Var}(X_L(f)) = \beta^2 \langle f, D_L f \rangle. \quad (7.15)$$

Thus D_L is a real symmetric positive-semidefinite covariance matrix. The exponential moment bound makes the MGF twice differentiable at zero; its value is one and its first derivative is zero. Subtract one from both sides of (7.12), divide by t^2 and take $t \rightarrow 0$. This gives $\text{Var}(X_L(f)) \leq 2K_L(f)$ and hence

$$\langle f, D_L f \rangle \leq \frac{1}{\beta c} \langle f, L_{\text{sp}}^{-1} f \rangle. \quad (7.16)$$

For $p = 2\pi k/L$, set $e_p(y) = V^{-1/2} e^{ip \cdot y}$ and $E(p) = \sum_{j=1}^3 (1 - \cos p_j)$. Directly from the nearest-neighbor Laplacian,

$$(L_{\text{sp}} e_p)(y) = \sum_{j=1}^3 \{2 - e^{ip_j} - e^{-ip_j}\} e_p(y) = 2E(p) e_p(y).$$

Spatial translation invariance diagonalizes D_L in the same basis; denote its eigenvalue by $\widehat{D}_L(p)$. The real inequality (7.16) extends to a complex field $f = a + ib$ by adding the real inequalities for a, b : symmetry cancels the imaginary cross terms in both Hermitian quadratic forms. Thus there is no doubling of the constant. Applying it to e_p for $p \neq 0$ yields

$$0 \leq \widehat{D}_L(p) \leq \frac{1}{2\beta c E(p)}. \quad (7.17)$$

No inverse or estimate is applied to the constant mode. This is a Fourier or quadratic-form upper bound, not an entrywise position-space comparison.

7.6 The singular three-dimensional sum and zero-mode subtraction

Define

$$I_3 = \frac{1}{(2\pi)^3} \int_{(-\pi, \pi]^3} \frac{d^3 p}{E(p)}, \quad I_{3,L} = \frac{1}{V} \sum_{p \neq 0} \frac{1}{E(p)}, \quad I_{3,L} \rightarrow I_3 < \infty. \quad (7.18)$$

Here is a proof of the asserted convergence. The elementary concavity bound $\sin t \geq 2t/\pi$ for $0 \leq t \leq \pi/2$ gives $1 - \cos p_j \geq 2p_j^2/\pi^2$, hence $E(p) \geq 2|p|^2/\pi^2$ on the centered Brillouin cube. Inclusion of $\{\|p\|_\infty \leq \delta\}$ in the radius- $\sqrt{3}\delta$ ball therefore gives the continuous tail bound

$$\frac{1}{(2\pi)^3} \int_{\|p\|_\infty \leq \delta} \frac{d^3 p}{E(p)} \leq \frac{\sqrt{3}\delta}{4} \quad (0 < \delta < \pi). \quad (7.19)$$

Choose centered integer representatives k for the discrete momenta. The shell $\|k\|_\infty = n$ has at most $(2n+1)^3 - (2n-1)^3 = 24n^2 + 2$ points, and $E(2\pi k/L) \geq 8|k|^2/L^2$. With $M = \lfloor \delta L/(2\pi) \rfloor$, we obtain

$$\frac{1}{V} \sum_{0 < \|p\|_\infty \leq \delta} \frac{1}{E(p)} \leq \frac{1}{8L} \sum_{n=1}^M \left(24 + \frac{2}{n^2} \right) \leq \frac{13M}{4L} \leq \frac{13\delta}{8\pi}. \quad (7.20)$$

An empty sum is zero. Outside the δ cube the integrand, with its cutoff, is bounded and Riemann integrable, so the ordinary grid sums converge there. First let even $L \rightarrow \infty$ for fixed δ , then let $\delta \downarrow 0$ using both tail bounds. This proves (7.18) without numerical quadrature or a decimal approximation. The controlling dimension is the spatial dimension three, not eight.

Finally the Fourier trace identity gives

$$\frac{\widehat{D}_L(0)}{V} = D_L(0, 0) - \frac{1}{V} \sum_{p \neq 0} \widehat{D}_L(p) \geq D_L(0, 0) - \frac{I_{3,L}}{2\beta c}. \quad (7.21)$$

The left side is nonnegative, but this section alone gives no strictly positive lower bound. A separate bound $D_L(0, 0) \geq d_\beta$ would imply $\liminf_L \widehat{D}_L(0)/V \geq d_\beta - I_3/(2\beta c)$. The collective and Falk–Bruch arguments in the next section must supply that input before any cusp or distinct-state conclusion is drawn.

8 Collective Duhamel lower bound

8.1 Thermal energy, translated forms and the collective moment

In this section L, β are fixed and the source is zero. Write $H = H_L(0) = -(2m)^{-1}\Delta + U$, with closed form h , and let (E_i, ψ_i) be an orthonormal eigenbasis. Such a basis exists because the finite heat trace makes the heat operator compact. Choose a constant $C = C_0$ large enough that, for some $\kappa > 0$, $U + C \geq 1 + \kappa \sum_y |q_y|^4$ as a form multiplier; the earlier envelope allows $\kappa = g/256$. Write $h_C = h + C$. The form domain in (2.3) is unchanged by this shift. For $s \geq 0$,

$$se^{-\beta s} \leq \frac{2}{e\beta} e^{-\beta s/2}, \quad \rho_L(H + C) < \infty, \quad \rho_L \left(\sum_y |q_y|^4 \right) < \infty. \quad (8.1)$$

The last two statements follow from the heat trace at $\beta/2$ and the form inequality $\kappa \sum_y |q_y|^4 \leq H + C$, respectively.

For later use we make the unbounded-integrability statement precise.

Lemma 8.1 (Thermal polynomial integrability). *At fixed L, β , if $P(q)$ is a multiplication polynomial of degree at most four, then the positive form trace*

$$\mathsf{T}_\beta(P) := \sum_i e^{-\beta E_i} \int_{\mathbb{R}^{8V}} |P(q)| |\psi_i(q)|^2 dq \quad (8.2)$$

is finite. In particular $\rho_L(|P|) < \infty$ and $\sum_i p_i |\langle \psi_i, P\psi_i \rangle| < \infty$.

Proof. In finite dimension, $|P(q)| \leq K_{P,L}(1 + w(q))$ for a constant depending only on the displayed polynomial and the fixed volume. The form inequality above gives

$$\int w |\psi_i|^2 dq \leq \kappa^{-1}(E_i + C)$$

for every normalized eigenvector. Since $E_i + C \geq 1$ and $\sum_i e^{-\beta E_i}(E_i + C) < \infty$ by the first inequality in (8.1) and the heat trace at $\beta/2$, summing the bound proves (8.2). The final statement follows from $|\langle \psi_i, P\psi_i \rangle| \leq \langle \psi_i, |P| \psi_i \rangle$. This is a positive form trace defined by the spectral sum, not a cyclic trace involving an unbounded product. $\square$

Smooth compactly supported functions are a form core: radial cutoffs converge in H^1 and the quartic weighted norm, with their gradient terms supported in escaping annuli; mollification then converges on each compact support, where the potential is bounded.

Let $v_{y,e} = u_e/\sqrt{V}$, so that $|v| = 1$, and set $H(t) = -(2m)^{-1}\Delta + U(q+tv)$ and $\Delta_t = U(q+tv) - U(q)$. Translation preserves H^1 and the quartic weighted space, using $|q_y - tv_y|^4 \leq 8(|q_y|^4 + |tv_y|^4)$. Thus the translated form has the same domain. The Hamiltonians are unitarily equivalent and $Z(t) = Z$. For fixed t , Δ_t has degree at most three in q and is absolutely Gibbs integrable. Put $p_i = e^{-\beta E_i}/Z$ and $d_i = \langle \psi_i, \Delta_t \psi_i \rangle$, interpreted as a multiplication form. Scalar Jensen applied to the spectral probability measure of $H(t)$ in ψ_i gives

$$\langle \psi_i, e^{-\beta H(t)} \psi_i \rangle \geq e^{-\beta h_t[\psi_i]} = e^{-\beta(E_i + d_i)}.$$

Only form-domain membership is needed for the spectral first moment. Summation and a second scalar Jensen inequality give

$$1 = \frac{Z(t)}{Z} \geq \sum_i p_i e^{-\beta d_i} \geq \exp\left(-\beta \sum_i p_i d_i\right). \quad (8.3)$$

The first bound also makes the intermediate exponential sum finite; $\sum_i p_i |d_i| < \infty$ follows from the polynomial bound. Consequently

$$\rho_L(U(q+tv) - U(q)) \geq 0. \quad (8.4)$$

The finite polynomial $t \mapsto \rho_L(\Delta_t)$ has a minimum at zero, so its second derivative at the minimum satisfies $\partial_t^2 \rho_L(\Delta_t)|_{t=0} \geq 0$. No assertion that the second derivative is nonnegative for every t is needed (and the negative quadratic coefficient $r < 0$ would not justify such an assertion). This differentiates finitely many integrable coefficients, not the heat trace under an unbounded cubic perturbation, and uses no operator monotonicity of the exponential. Spatial differences and internal differences are unchanged by the common displacement. The mass term contributes r , the scalar quartic contributes $3g \sum_y S_y/(8V)$, and each locking edge contributes $\lambda(q_{y,e} - q_{y,f})^2/(8V)$: the second derivative of $q_{y,e}^2$ is $4/(8V)$. Hence

$$B_L := \partial_t^2 U(q+tv)|_{t=0} = r + \frac{3g}{8V} \sum_y S_y + \frac{\lambda}{8V} \sum_y D_y^{\text{int}}, \quad \rho_L(B_L) \geq 0. \quad (8.5)$$

For $\Pi_0 = -i\hbar v \cdot \nabla = V^{-1/2} \sum_y (u, p_y)$, direct differentiation on the smooth compact core gives

$$[\Pi_0, [H_L(0), \Pi_0]] = \hbar^2 B_L. \quad (8.6)$$

The right side is a quadratic, form-bounded multiplier and defines the continuous form extension of this core identity. No Gibbs trace of a product of three unbounded operators is used: positivity of its expectation was established separately in (8.3). The $\hbar^2$ belongs to the momentum commutator, not the displacement Hessian. In particular, (8.5) is not pointwise positivity of B_L .

Spatial translation invariance gives the first inequality below. At zero source parity makes each coordinate mean zero, and the actual loop FKG and clipping argument in (6.5) makes all off-diagonal coordinate products nonnegative in expectation. Therefore

$$-r \leq \frac{3g}{8} \rho_L(S_0) + \frac{\lambda}{8} \rho_L(D_0^{\text{int}}), \quad \rho_L(Q_0^2) \geq \frac{1}{8} \rho_L(S_0). \quad (8.7)$$

Indeed, the second inequality follows by expanding the square of $Q_0 = \sum_e q_{0,e}/\sqrt{8}$. The three-regular internal graph gives

$$\rho_L(D_0^{\text{int}}) = 3\rho_L(S_0) - 2 \sum_{\{e,f\} \in E(Q_3)} \rho_L(q_{0,e} q_{0,f}) \leq 3\rho_L(S_0). \quad (8.8)$$

These are Gibbs expectation inequalities, not pointwise graph bounds. Together, for $r < 0$, they imply

$$\rho_L(Q_0^2) \geq \theta_Q, \quad \theta_Q = -\frac{r}{3(g+\lambda)} > 0. \quad (8.9)$$

8.2 A finite spectral proof of the Falk–Bruch inequality

We give the scalar operator inequality used here explicitly; it is the standard Falk–Bruch inequality, not a new model-specific result. Its normalization agrees with [5, Proposition 3.18, (3.65), (3.68)]. For a bounded self-adjoint A , define

$$g(A) = \sum_{i,j} p_i |A_{ij}|^2 = \frac{1}{2} \sum_{i,j} (p_i + p_j) |A_{ij}|^2,$$

$$b(A) = \sum_{i,j} \mathcal{L}(p_i, p_j) |A_{ij}|^2, \quad \mathcal{L}(a, b) = \int_0^1 a^{1-t} b^t dt = \begin{cases} (a-b)/(\log a - \log b), & a \neq b, \\ a, & a = b. \end{cases} \quad (8.10)$$

The integral and weighted arithmetic–geometric mean imply $0 < \mathcal{L}(a, b) \leq (a+b)/2$. Thus $0 \leq b(A) \leq g(A)$, and polarization gives a positive form with Cauchy–Schwarz. Spectral expansion and Tonelli identify

$$b(A) = \frac{1}{\beta Z} \int_0^\beta \text{Tr} \left(e^{-(\beta-\tau)H} A e^{-\tau H} A \right) d\tau. \quad (8.11)$$

This trace is the nonnegative spectral sum, equivalently the squared Hilbert–Schmidt norm of $e^{-\tau H/2} A e^{-(\beta-\tau)H/2}$; it is finite, including at the endpoints. For $0 < \tau < \beta$, the elementary bound $p_i^{1-\tau/\beta} p_j^{\tau/\beta} \leq (1-\tau/\beta)p_i + (\tau/\beta)p_j$ suffices. No separately defined positive-time exponential $e^{+\tau H}$ is needed. For bounded multipliers this is also the finite-volume loop two-time integral.

In a finite-dimensional Gibbs system set

$$c(A) = \beta \sum_{i,j} (E_j - E_i) (p_i - p_j) |A_{ij}|^2 = \rho([A, [\beta H, A]]) \geq 0. \quad (8.12)$$

Let $f(0) = 1$ and, for $k > 0$, let $x > 0$ be the unique solution of $x \tanh x = k$ and define $f(k) = \tanh x/x$. The function $x \tanh x$ increases continuously from zero to infinity, so f is well-defined and continuous on $[0, \infty)$.

For completeness, the scalar function $\Phi(u) = \sqrt{u} \coth \sqrt{u}$, with $\Phi(0) = 1$, is increasing and concave. For $x > 0$ differentiation gives

$$\Phi'(x^2) = \frac{\sinh x \cosh x - x}{2x \sinh^2 x} > 0,$$

$$\Phi''(x^2) = \frac{\cosh x}{4x^3 \sinh^3 x} (2x^2 - x \tanh x - \sinh^2 x) < 0. \quad (8.13)$$

The first numerator starts at zero and has derivative $2 \sinh^2 x > 0$. For the second sign, $0 \leq \tanh x \leq x$ and $(\tanh x - x + x^3/3)' = x^2 - \tanh^2 x \geq 0$ imply $\tanh x \geq x - x^3/3$. Taylor’s formula with nonnegative remainder gives $\sinh x \geq x + x^3/6$. Consequently $\sinh^2 x + x \tanh x \geq 2x^2 + x^6/36$. Continuity at zero extends the concavity to the closed half-line.

For a nonzero self-adjoint finite matrix A , $b = b(A) > 0$. Put $x_{ij} = |\log p_i - \log p_j|/2$ and $w_{ij} = \mathcal{L}(p_i, p_j) |A_{ij}|^2/b$. These weights sum to one, including any degenerate-energy pairs with $x_{ij} = 0$. The exact identities

$$\frac{g}{b} = \sum_{i,j} w_{ij} \Phi(x_{ij}^2), \quad \frac{c}{4b} = \sum_{i,j} w_{ij} x_{ij}^2 \quad (8.14)$$

follow from the arithmetic/logarithmic mean ratio $x \coth x$ and $\beta(E_j - E_i) = \log p_i - \log p_j$. Jensen now gives $g/b \leq \Phi(c/(4b))$. If $c = 0$, every active pair has zero energy gap and $g = b$. If $c > 0$, set $z = \sqrt{c/(4b)}$ to obtain $g \leq c/(4z \tanh z)$. If $x \tanh x = c/(4g)$, monotonicity gives $x \geq z$, and hence

$$b(A) \geq g(A) f \left(\frac{c(A)}{4g(A)} \right). \quad (8.15)$$

For $A = 0$ interpret the right side as zero. This proves the precise finite inequality needed below without importing a radial moment or phase theorem.

8.3 Bounded coordinates, energy-weighted sums and cutoff removal

Return to the fixed finite-volume Schrödinger Hamiltonian. Let $A_R = R \tanh(Q_0/R)$ for $R > 0$. At fixed R , both A_R and A_R^2 are bounded multipliers with bounded first derivatives and hence preserve the closed form domain. Since $\|u\| = 1$, $|\nabla A_R|^2 = \text{sech}^4(Q_0/R) \leq 1$. On the smooth form core, the potential and mixed gradient terms cancel in

$$h[A_R\psi] - \text{Re } h(\psi, A_R^2\psi) = \frac{1}{2m} \int |\nabla A_R|^2 |\psi|^2 dq.$$

Form density passes the identity to the domain. For an eigenvector, $A_R^2\psi_i$ is an admissible form test, so the eigenvector form equation gives

$$h[A_R\psi_i] - E_i \|A_R\psi_i\|^2 = \frac{1}{2m} \int \text{sech}^4(Q_0/R) |\psi_i|^2 dq. \quad (8.16)$$

Before regrouping any energy-weighted sums, observe that $h_C[A_R\psi] \leq C_R(h_C[\psi] + \|\psi\|^2)$, by the product rule, $U + C \geq 0$, and the multiplier bounds. Thus

$$\begin{aligned} \sum_{i,j} p_i(E_j + C) |(A_R)_{ij}|^2 &= \sum_i p_i h_C[A_R\psi_i] < \infty, \\ \sum_{i,j} p_i(E_i + C) |(A_R)_{ij}|^2 &\leq R^2 \rho_L(H + C) < \infty. \end{aligned} \quad (8.17)$$

Here self-adjointness allows the row/column interchange for the first identity. Since $E_i + C \geq 0$, these bounds control $\sum_{i,j} p_i |E_j - E_i| |(A_R)_{ij}|^2$; the corresponding p_j sum is equal after interchanging indices. Regrouping is therefore legitimate, and (8.16) gives the finite scalar

$$\begin{aligned} c_R &:= \beta \sum_{i,j} (E_j - E_i)(p_i - p_j) |(A_R)_{ij}|^2 \\ &= 2\beta \sum_i p_i (h[A_R\psi_i] - E_i \|A_R\psi_i\|^2) = \frac{\beta}{m} \rho_L(\text{sech}^4(Q_0/R)), \quad 0 \leq c_R \leq \frac{\beta}{m}. \end{aligned} \quad (8.18)$$

Every paired spectral summand is nonnegative. This is the form interpretation of $\rho_L([A_R, [\beta H, A_R]])$, not a formal cyclic trace calculation. The scalar c_R is the expectation of the displayed bounded multiplier; the two objects are not identified.

To apply (8.15), let P_M project onto the first M eigenvectors, $H_M = \text{diag}(E_1, \dots, E_M)$, $A_{R,M} = P_M A_R P_M$, $Z_M = \sum_{i \leq M} e^{-\beta E_i}$ and $q_M = Z_M/Z$. The finite Gibbs probabilities are p_i/q_M . Let $g_{R,M}, b_{R,M}, c_{R,M}$ be the three finite expressions. Homogeneity of the logarithmic mean gives

$$\begin{aligned} q_M g_{R,M} &= \sum_{i,j \leq M} p_i |(A_R)_{ij}|^2, \\ q_M b_{R,M} &= \sum_{i,j \leq M} \mathcal{L}(p_i, p_j) |(A_R)_{ij}|^2, \\ q_M c_{R,M} &= \beta \sum_{i,j \leq M} (E_j - E_i)(p_i - p_j) |(A_R)_{ij}|^2. \end{aligned} \quad (8.19)$$

The unnormalized right sides increase to $g(A_R), b(A_R), c_R$ and $q_M \rightarrow 1$. Thus all three normalized quantities converge at fixed R, L, β . Since A_R is nonzero and the Gibbs state is faithful, $g(A_R) > 0$ and the finite $g_{R,M}$ are eventually positive. Continuity of f passes (8.15) to

$$b(A_R) \geq g(A_R) f\left(\frac{c_R}{4g(A_R)}\right). \quad (8.20)$$

We do not assert monotonicity of the normalized finite expressions, operator-norm convergence of commutators, or an exact canonical commutation relation in a finite matrix algebra.

Only now let $R \rightarrow \infty$, still at fixed L, β . Since $|A_R| \leq |Q_0|$, $A_R \rightarrow Q_0$ and $\rho_L(Q_0^2) < \infty$, dominated convergence gives

$$g(A_R) \rightarrow s_L := \rho_L(Q_0^2), \quad g(A_R - Q_0) \rightarrow 0, \quad c_R \rightarrow \frac{\beta}{m}. \quad (8.21)$$

For the last limit use bounded convergence in (8.18). The bound $b \leq g$ and Cauchy–Schwarz extend b to the thermal L^2 completion and give $|\sqrt{b(A_R)} - \sqrt{b(Q_0)}| \leq \sqrt{g(A_R - Q_0)}$. There is also direct loop identification: time invariance and Cauchy–Schwarz bound the difference of the two-time products uniformly in τ by

$$\mathbb{E}_{L,0}|A_R(\tau)A_R(0) - Q_0(\tau)Q_0(0)| \leq (\sqrt{g(A_R)} + \sqrt{s_L})\sqrt{g(A_R - Q_0)}. \quad (8.22)$$

Here $A_R(\tau)$ denotes the clipped loop coordinate. Integration proves $b(Q_0) = D_L(0, 0)$ without positing an unbounded operator trace. Parity makes both odd means zero, so raw and connected forms agree. Therefore

$$D_L(0, 0) \geq s_L f\left(\frac{\beta}{4ms_L}\right), \quad s_L = \rho_L(Q_0^2) \geq \theta_Q > 0. \quad (8.23)$$

The local coefficient is $\beta/m = \beta\hbar^2/\chi$: it contains the β from βH and is distinct from the global momentum identity (8.6). To substitute the moment lower bound, fix $k > 0$ and write $x_s \tanh x_s = k/s$. Then $sf(k/s) = k/x_s^2$. Increasing s decreases x_s , so this expression increases. Using $k = \beta/(4m)$ yields

$$D_L(0, 0) \geq d_\beta := \theta_Q \frac{\tanh x_\beta}{x_\beta}, \quad x_\beta \tanh x_\beta = \frac{\beta}{4m\theta_Q}. \quad (8.24)$$

All domain arguments were at fixed volume: finite spectral cutoff first, $M \rightarrow \infty$ at fixed R , then $R \rightarrow \infty$. The final scalar bound has no V and can now be combined with the infrared sum before taking the spatial limit. It constructs no common infinite-volume unbounded operator, nor a volume-uniform operator core.

9 Strict cusp and two DLR states

9.1 The two proof branches and their common normalization

Fix $m, c, g, \lambda > 0$ and $r < 0$, and keep one $\beta > 0$ throughout the spatial and source limits. The zero-source fluctuation estimate and the positive-source DLR construction use the same physical Hamiltonian, but different sequences of laws. Their common dictionary is

$$\frac{Z_L(h)}{Z_L(0)} = \mathbb{E}_{L,0} e^{hX_L}, \quad Y_L := \frac{X_L}{V}, \quad \mathbb{E}_{L,0} Y_L^2 = \beta^2 \Pi_L, \quad P_\beta = \frac{p_\beta}{8\beta}. \quad (9.1)$$

Here X_L already contains the time integral in (3.20). In particular, the source in the exponential is h , not βh . The moment estimate below is in $\mu_{L,0}$; the tangent states later are limits of μ_{L,h_j} with h_j held fixed during each volume limit. No zero-source periodic accumulation is asserted to be a positive phase.

Time invariance, (7.15), and the Fourier trace identity give

$$\begin{aligned} \Pi_L &:= \mathbb{E}_{L,0} \left[\left(\frac{X_L}{\beta V} \right)^2 \right] = \frac{1}{V^2} \sum_{y,z} D_L(y, z) = \frac{\hat{D}_L(0)}{V} \\ &= D_L(0, 0) - \frac{1}{V} \sum_{p \neq 0} \hat{D}_L(p). \end{aligned} \quad (9.2)$$

Thus the positive local bound is used only after subtracting every nonzero mode. Combining (7.17), (7.18), and (8.24) first gives $\Pi_L \geq d_\beta - I_{3,L}/(2\beta c)$ and then

$$\liminf_{L \rightarrow \infty} \Pi_L \geq \delta_\beta := d_\beta - \frac{I_3}{2\beta c}. \quad (9.3)$$

Define

$$A_0 = 8cm\theta_Q^2. \quad (9.4)$$

Because $\beta = 4m\theta_Q x_\beta \tanh x_\beta$,

$$2\beta c \delta_\beta = A_0 \tanh^2 x_\beta - I_3. \quad (9.5)$$

If $A_0 > I_3$, set

$$\rho_* = \sqrt{I_3/A_0}, \quad x_* = \operatorname{artanh}(\rho_*), \quad \beta_* = 4m\theta_Q x_* \rho_*. \quad (9.6)$$

Then $\beta > \beta_*$ implies $\delta_\beta > 0$. At equality, or when $A_0 \leq I_3$, the lower bound is nonpositive; this is not a phase-absence statement. Indeed $x \tanh x$ and $\tanh^2 x$ are strictly increasing for $x > 0$, so for $A_0 > I_3$ the condition $\delta_\beta > 0$ is equivalent to $\beta > \beta_*$. Also $0 < I_3 < \infty$: positivity follows from the positive integrand away from the origin, and finiteness from (7.19). The sufficient set is nonempty. For fixed $c, g, \lambda > 0$ and $r < 0$, choose $m > I_3/(8c\theta_Q^2)$ and then $\beta > \beta_*$. This is a choice of finite parameters, not a zero-temperature or infinite-mass limit. The threshold is sufficient and is not identified with the exact critical temperature.

9.2 A squared-tail estimate at a nondifferentiable pressure

We use the following elementary endpoint lemma rather than differentiating finite-volume second moments through the cusp.

Lemma 9.1 (Even-pressure squared-tail bound). *Let $V_n > 0$ tend to infinity and let X_n have finite moment generating functions on a common interval $(-h_0, h_0)$, $h_0 > 0$. Suppose*

$$F_n(h) = V_n^{-1} \log \mathbb{E} e^{hX_n} \longrightarrow F(h)$$

pointwise on that interval, where F is finite, even, and $F(0) = 0$. Then F is convex, its right derivative $\ell = F'_+(0)$ is finite and nonnegative, and

$$\limsup_n \mathbb{E}[(X_n/V_n)^2] \leq \ell^2. \quad (9.7)$$

Proof. Convexity passes from the log MGFs to their finite limit. Evenness makes zero a minimum, and finiteness on a two-sided interval bounds the one-sided slopes there. Fix $\varepsilon > 0$ and choose a fixed $h \in (0, h_0)$ with $F(h)/h \leq \ell + \varepsilon/4$. Pointwise convergence and evenness give, for large n , $\max(F_n(h), F_n(-h)) \leq h(\ell + \varepsilon/2)$. Chernoff's bound for $Y_n = X_n/V_n$ gives

$$\Pr(|Y_n| > z) \leq 2e^{-V_n h(z - \ell - \varepsilon/2)} \quad (z \geq \ell + \varepsilon).$$

For $R = \ell + \varepsilon$, the exact layer-cake identity and Tonelli give

$$\begin{aligned} \mathbb{E}[Y_n^2 1_{\{|Y_n| > R\}}] &= R^2 \Pr(|Y_n| > R) + \int_R^\infty 2z \Pr(|Y_n| > z) dz \\ &\leq 2e^{-V_n h\varepsilon/2} \left[R^2 + \frac{2R}{V_n h} + \frac{2}{(V_n h)^2} \right] \longrightarrow 0. \end{aligned} \quad (9.8)$$

The central contribution is at most R^2 . Take the limsup and then $\varepsilon \downarrow 0$. $\square$

This is the needed second-moment specialization of [5, Proposition 3.9, (3.23)–(3.24)], proved here using only a neighborhood of zero. It uses fixed nonzero test sources before taking $V_n \rightarrow \infty$, and controls a squared tail rather than merely weak convergence or tail probabilities. No derivative at zero is interchanged with the volume limit.

Apply the lemma to the scalar pushforward of $\mu_{L,0}$ by X_L . Equation (9.1) gives $F_L(h) = p_{\beta,L}(h) - p_{\beta,L}(0)$, which converges to the finite, even $F(h) = p_\beta(h) - p_\beta(0)$ by (4.7). For $\delta_\beta > 0$, the two independently justified bounds give

$$\beta^2 \delta_\beta \leq \liminf_L \mathbb{E} Y_L^2 \leq \limsup_L \mathbb{E} Y_L^2 \leq [p'_\beta(0+)]^2. \quad (9.9)$$

The right slope is nonnegative and finite, so taking its positive square root and dividing by 8β yields

$$p'_\beta(0+) \geq \beta\sqrt{\delta_\beta}, \quad D_+P_\beta(0) = \frac{p'_\beta(0+)}{8\beta} \geq \frac{\sqrt{\delta_\beta}}{8} > 0. \quad (9.10)$$

Evenness gives $D_-P_\beta(0) = -D_+P_\beta(0)$, so the pressure has a strict collective-source cusp in the sufficient regime. At every finite volume, in contrast, $p'_{\beta,L}(0) = 0$ by parity. The derivative convergence (4.8) applies only at differentiability points of the limiting pressure and makes no assertion at this cusp.

9.3 Source-selected DLR states and a bounded witness of distinctness

Choose positive differentiability points $h_j \downarrow 0$. Equations (5.20) and (9.10) give a zero-source DLR state μ_+ with

$$\mu_+(Q_0) = 8D_+P_\beta(0) \geq \sqrt{\delta_\beta} > 0. \quad (9.11)$$

For clarity, this construction uses both passages in Section 5.5: at each fixed h_j , take a spatial accumulation and identify its clipped local expectation with $8P'_\beta(h_j)$; then take a source-to-zero subsequence using common W_t tightness, compact-uniform kernel continuity and the second clipping passage. A cusp by itself would not establish the DLR property.

Global parity $\Theta\omega = -\omega$ is a continuous involution of Ω_t . The Gaussian reference is symmetric and $I_\Delta^0(-\omega \mid -\xi) = I_\Delta^0(\omega \mid \xi)$, so change of variables in the finite specification gives, for bounded Borel f ,

$$\pi_\Delta^0(f \mid \Theta\xi) = \pi_\Delta^0(f \circ \Theta \mid \xi). \quad (9.12)$$

Consequently $(\Theta_*\mu_+)\pi_\Delta^0(f) = \mu_+\pi_\Delta^0(f \circ \Theta) = \mu_+(f \circ \Theta)$. Thus $\mu_- := \Theta_*\mu_+$ is also a tempered DLR state and

$$\mu_-(Q_0) = -\mu_+(Q_0) < 0. \quad (9.13)$$

They are distinct because their finite local expectations differ. One can also exhibit a bounded continuous cylinder witness. Let $a_\beta = \sqrt{\delta_\beta}$ and $C_2 = C_{\text{per}}/\ell_\sigma$ from (5.18). Parity preserves that second-moment bound. For $f_R = \text{clip}_R(Q_0)$, the two clipping errors give

$$\mu_+(f_R) - \mu_-(f_R) \geq 2a_\beta - \frac{2C_2}{R} > a_\beta > 0 \quad \text{whenever } R > \frac{2C_2}{a_\beta}. \quad (9.14)$$

This uses a finite R after the states have been constructed. They are not asserted to be extremal, pure, clustering, or the only phases.

Theorem 9.2 (Conditional Q3LOCK phase-coexistence package). *Fix the Hamiltonian (2.2), with $m, c, g, \lambda > 0$ and $r \in \mathbb{R}$, on even periodic cubes in the fixed lattice $\mathbb{Z}^3$ and use the pressure and source normalization (3.20). Assume the finite-volume mesh/loop identification, the KP general-vector Feller and compactness hypotheses, the selected-law FKG passage, the finite FSS inequality, the collective form/Falk–Bruch passage, and the source-window DLR continuity described in Sections 3–8, with the displayed uniformities and limit orders. These are upstream analytic inputs, not an assumption of a cusp or state multiplicity. The pressure limit is finite, even and convex for every real source, and the tempered DLR set at each fixed source is nonempty and W_t -compact. If also $r < 0$, $A_0 > I_3$, and $\beta > \beta_*$, then the pressure has the strict cusp (9.10); moreover there are at least two distinct parity-related tempered Euclidean DLR states at $h = 0$ satisfying (9.11)–(9.13).*

Proof. The finite loop/trace and source identities are established in Section 3. The two-sided pressure bounds, open-box limit and periodic seam estimate in Section 4 yield the common finite pressure; the general-vector input in Section 5.1 gives fixed-source DLR existence and compactness. At zero source, actual-loop FKG in Section 6 enters the collective moment bound, while Section 8 turns it into $D_L(0,0) \geq d_\beta$. The nonzero-mode and singular-sum arguments in Section 7 therefore yield (9.3). The strict threshold algebra and Lemma 9.1, with the exact scalar source pushforward, give (9.9) and the cusp. The separate source-window DLR/tangent construction supplies μ_+ ; (9.12) supplies μ_- , and (9.14) distinguishes them. Each implication uses the same fixed β and physical coefficients. No simultaneous $h = h(L)$ limit or information about all DLR mixtures is required. $\square$

Remark 9.3 (Status of the theorem). The theorem is the target conditional composition of the registered proof blocks. Detailed arguments and a source-to-conclusion composition are now present, but their presence and finite diagnostics do not constitute signed independent acceptance. The current repository tier is T0 and the theorem must be reopened if any source convention, limit order, domain condition, or normalization fails independent audit.

10 Proof dependency and literature crosswalk

Table 1 separates reused standard machinery from the model-specific interfaces. Exact bibliographic details and source hashes are in the companion `literature-crosswalk.md`.

Source or input	Role	Boundary in this paper
Simon, arXiv:math-ph/9907022v1	Free-bridge formula, Theorem 1.1	Harmonic reweighting and absolute normalization are derived here
Kozitsky–Pasurek, arXiv:math-ph/0609045v1	General-vector DLR existence/compactness and Gaussian Fernique input	Feller/source-window passages are written here; no scalar phase theorem is imported
Froehlich–Simon–Spencer (1976)	Finite-dimensional Poisson-shift infrared inequality	Applied at fixed time mesh after an explicit $8N$ -component source map; loop passage is separate
Fortuin–Kasteleyn–Ginibre (1971)	Finite partially ordered association mechanism	Continuous-loop association is proved by mesh induction and weak-limit passage
Kargol–Kondratiev–Kozitsky, arXiv:0710.2303v1	Endpoint Griffiths and Falk–Bruch comparators	The endpoint specialization and finite spectral inequality are proved here; no radial theorem is imported
Kozitsky, LMP 51 (2000); CMP 5 (2002) [7, 8]	Vector/isotropic quantum-stabilization and normal-fluctuation comparators	Their radial/isotropic bounds do not supply the non-radial Q3LOCK cusp or DLR pair; no result is imported
Schneider–Beck–Stoll [11]	Radial n -component structural-transition comparison; Eq. (11) is stated for $n = 1, 2, \infty$	Comparison-only; the radial model, explicit n -range, and displacive/large- n conclusion do not cover non-radial finite- $n=8$ Q3LOCK
EXP-001587/588/589	Corrected residual, finite pressure, and open/periodic pressure limit	Internal model-specific inputs, still awaiting signed audit
EXP-001591/593/595/1598	DLR tangent, FKG, infrared, and collective lower-bound blocks	Conditional composition; finite diagnostics are not proof certification
Faris–Minlos [9]	Multidimensional anharmonic oscillators and a small-coupling ground-state cluster expansion	Zero-temperature ground-state comparison only; no finite-temperature DLR, source cusp, or parity-state result is imported
Froehlich–Lieb [10]	Anisotropic finite-spin classical/quantum phase-transition methods in two dimensions	Comparison-only; no direct unbounded continuous-loop, three-dimensional DLR, or Q3LOCK cusp theorem is imported

Table 1: Proof dependency and source boundary.

The bounded primary supplement [9, 10] is included to make the non-subsumption boundary explicit. Faris–Minlos is a small-coupling zero-temperature multidimensional-oscillator comparison, whereas Froehlich–Lieb concerns finitely degenerate anisotropic spin systems on a two-dimensional lattice. Neither source supplies the finite-temperature unbounded-loop/DLR/source-cusp/parity composition proved conditionally here. The comparison is not a novelty or priority claim; specialist literature review remains open. Schneider–Beck–Stoll [11] is checked directly as a radial n -component structural-transition source. Its quantum inequality is stated only for $n = 1, 2, \infty$, and its conclusions concern zero-point suppression and large- n critical behaviour rather than the present finite-temperature non-radial $n=8$ DLR/cusp package. It is therefore a comparison-only boundary, not an imported theorem.

FSS Theorem 2.1 already permits nonradial priors: lack of internal rotation symmetry is not a new infrared theorem. Likewise the functional form of our sufficient condition is standard. With $f(x \tanh x) = \tanh x/x$,

$$d_\beta = \theta_Q f(\beta/(4m\theta_Q)), \quad 2\beta c\theta_Q f(\beta/(4m\theta_Q)) > I_3. \quad (10.1)$$

This is the same scalar threshold mechanism as KKK [5], Theorem 3.20, (3.71)–(3.75), after replacing its model-dependent lower-bound parameter by θ_Q . This substitution compares formulas; it does not validate that parameter for Q3LOCK. That requires Section 8. The coefficient eight in A_0 arises here from the factor two in the infrared subtraction and the factor four in the Falk–Bruch argument, not from counting the eight internal coordinates.

The precise obstruction to directly importing KKK Theorems 3.20–3.21 is their radial-potential assumption, not the spatial lattice. Let W denote the quartic part of B_0 , and let e_0 be a coordinate unit vector. The vectors e_0 and u both have norm one, but

$$W(e_0) = \frac{g + 3\lambda}{4}, \quad W(u) = \frac{g}{32}, \quad W(e_0) - W(u) = \frac{7g}{32} + \frac{3\lambda}{4} > 0. \quad (10.2)$$

Only the three edges incident to e_0 contribute in the first evaluation; all locking differences vanish in the second. Thus even an orthogonal change of coordinates cannot make this potential radial.

Flattening (y, e) into scalar sites does not give the usual scalar model with only onsite potentials and bilinear pair couplings. An internal edge contains

$$\frac{\lambda}{4}(x - y)^2(x^2 + y^2) = \frac{\lambda}{4}(x^4 + y^4 - 2x^3y + 2x^2y^2 - 2xy^3). \quad (10.3)$$

Its mixed derivative is $\lambda(-3x^2/2 + 2xy - 3y^2/2)$, not a constant bilinear coupling. This rules out that direct rewriting, not all possible comparison theorems or reductions. Setting $\lambda = 0$ instead gives eight decoupled scalar crystals and changes the model; restricting every q_y to the line spanned by u also changes the finite-volume law.

The scalar asymmetric-potential result [6, Theorem 1.4] is an established one-component comparator: it permits a polarization discontinuity at some, possibly nonzero, source for a scalar crystal in $d \geq 3$ when mass and interaction are sufficiently large. The 2008 KKK vector theorem instead assumes rotation-invariant onsite structure, while its asymmetric results are scalar. Neither source supplies the present non-radial R^8 Q3LOCK block, the positive- λ collective lower bound, the zero-source strict cusp, or the parity-related tempered DLR pair with the bounded witness. Setting $\lambda = 0$ or restricting every vector to the collective line changes the finite-volume law and is not a reduction. The candidate contribution is consequently the exact positive- λ collective lower bound and its explicit loop/DLR composition, not the standard inequalities or threshold shape. This comparison narrows a direct-import shortcut but does not establish novelty or publication value; a broader anisotropic theorem and specialist assessment remain to be checked.

11 Reproducibility and adversarial boundary

The paper is accompanied by the canonical R-497 source manifest and the following deterministic finite diagnostics:

Command	Registered result
<code>q3lock_absolute_partition_audit.py</code>	EXP-001588, exact determinant
<code>q3lock_thermodynamic_pressure_audit.py</code>	EXP-001589, seam/tiling
<code>q3lock_dlr_tangent_content_audit.py</code>	EXP-001591, 70/70
<code>q3lock_fkg_content_audit.py</code>	EXP-001593, 504/504
<code>q3lock_reflection_infrared_content_audit.py</code>	EXP-001595, finite graph/source checks
<code>q3lock_collective_falk_bruch_content_audit.py</code>	EXP-001598, 40/40
<code>q3lock_strict_cusp_tangent_content_audit.py</code>	EXP-001598, 42/42

The exact paths, run hashes, and expected outputs are in `verification/README.md`. Each diagnostic was replayed twice with byte-identical output, but the diagnostics test finite algebra and provenance; they do not replace a mathematical proof or an external referee.

The adversarial checks that must be signed are:

1. the corrected seam coefficient 288 and all source/ β factors;
2. form-domain membership of the bounded coordinate and the order of finite spectral cutoff, coordinate cutoff, and spatial limit;
3. the typed FSS Poisson norm, including the absence of an extra factor of eight and the exclusion of the zero mode;
4. transfer of FKG association and fourth-moment products through the actual weighted loop limit;
5. the squared-tail estimate used at the nondifferentiable pressure point;
6. compact-uniform source continuity and the clipped expectation passage from $h_j \downarrow 0$ to a genuine DLR state;
7. the exact sufficient regime, with no inference at equality or outside it.

12 Readiness gates and conclusion

The present source is organized for review, not declared ready for submission. The release conditions are:

- (G1) complete remaining source-hypothesis/content review and any resulting repairs, followed by independent line-by-line mathematical acceptance of all seven blocks;
- (G2) exact literature hypothesis crosswalk and specialist novelty opinion;
- (G3) bounded claim/result-lineage decision consistent with R-497's T0 scope;
- (G4) final notation, bibliography, nonclaim, and convention freeze;
- (G5) fresh source-hash capture and primary/independent/integrated clean snapshot replay with byte-stable outputs;
- (G6) release check on the frozen tree and operator confirmation of commit and backup;
- (G7) after any resulting revision, rebuild and visually inspect the final submission PDF. This earlier review PDF is not that approval.

No gate above is silently discharged by the local finite diagnostics. The operator authorized this review edition on 2026-09-08 to enable external assessment before those gates close. The correct description is *conditional manuscript supplied for independent review; external acceptance and final submission remain open*.

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